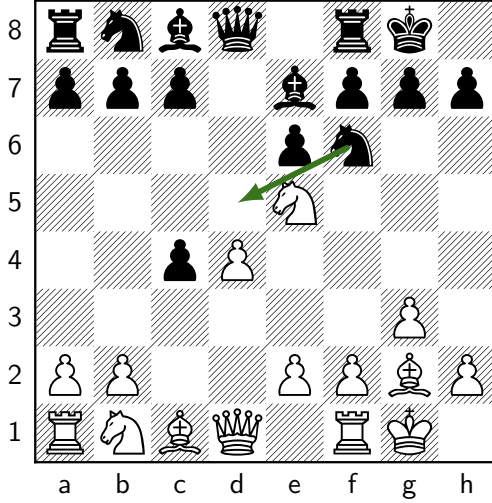


Melanie-BASE — Catalan

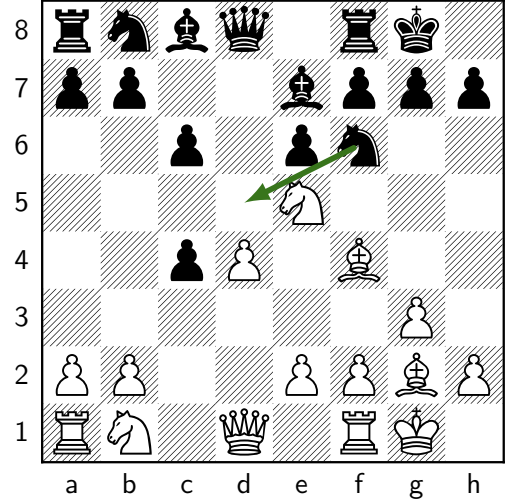
Melanie (Deep_Vision)

Abstract

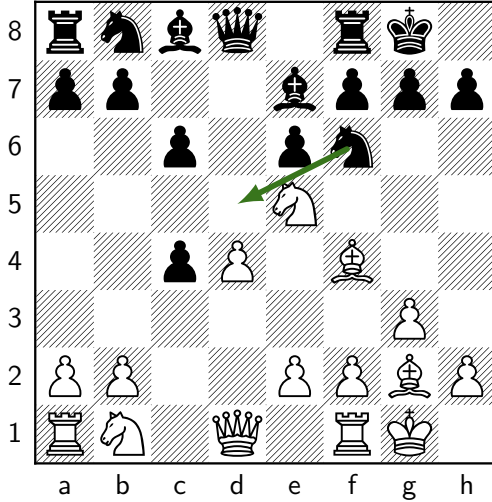
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



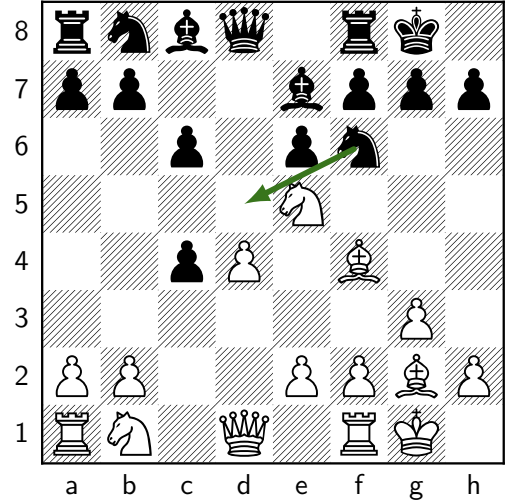
(a)



(b)

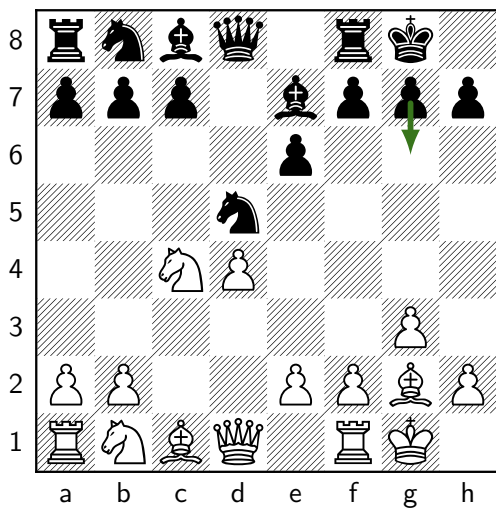


(c)

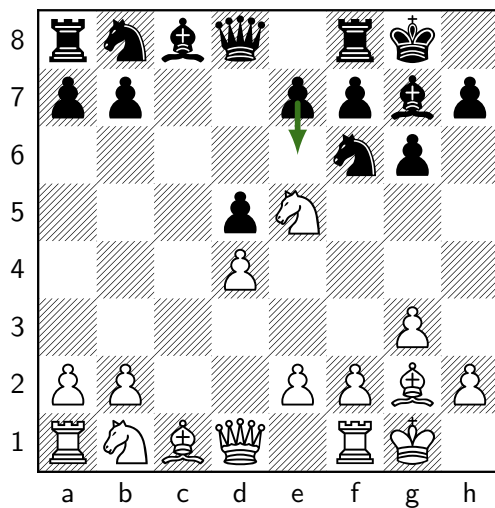


(d)

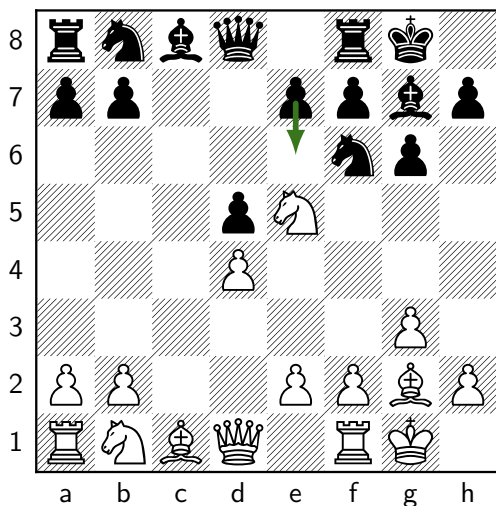
Figure 1: **Move 7, Black: Nd5.** (a) Our position; Melanie plays Nd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#271 (ply 17) [structural distance $d=26$, QAP obj=146]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#272 (ply 17) [structural distance $d=26$, QAP obj=146]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3736 (ply 17) [structural distance $d=26$, QAP obj=146]: the master played Nd5; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $ne5$ as their $bg2$; transferred move f6d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd5 (summed QAP objective 875, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



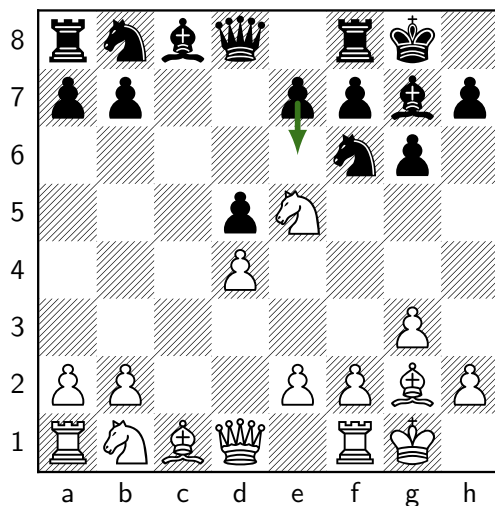
(a)



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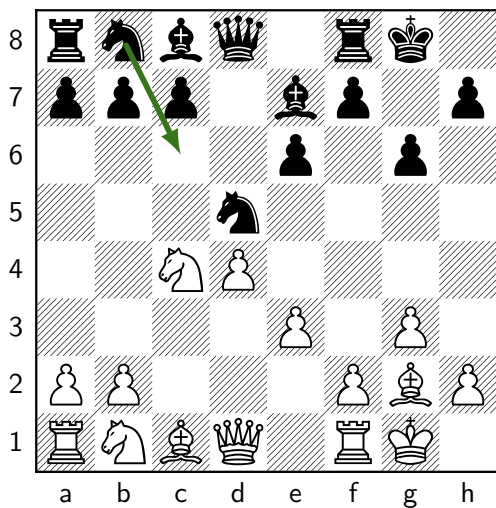


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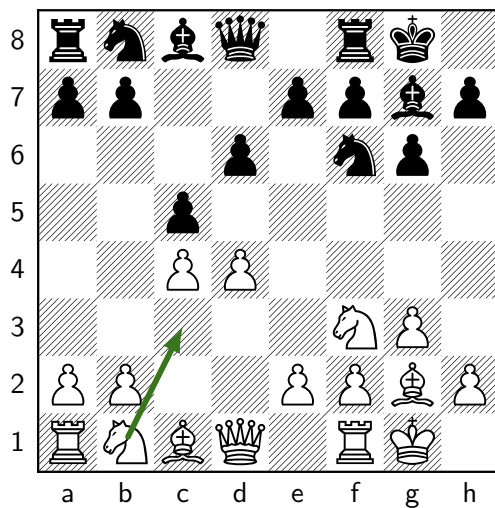


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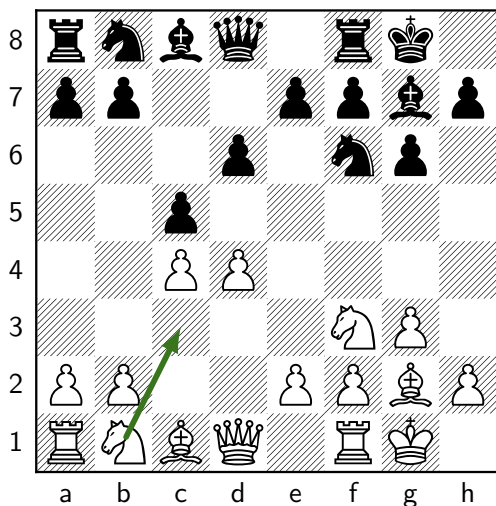
Figure 2: **Move 8, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `crr1_top.p0001#88` (ply 15) [structural distance $d=48$, QAP obj=20]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Be7$ as their $Ra8$, our $Pb7$ as their $Nb8$; transferred move g7g6 (**endorses this move**). (c) Retrieved from `crr1_top.p0001#89` (ply 15) [structural distance $d=48$, QAP obj=20]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Be7$ as their $Ra8$, our $Pb7$ as their $Nb8$; transferred move g7g6 (**endorses this move**). (d) Retrieved from `crr1_top.p0003#2420` (ply 15) [structural distance $d=48$, QAP obj=20]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Be7$ as their $Ra8$, our $Pb7$ as their $Nb8$; transferred move g7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



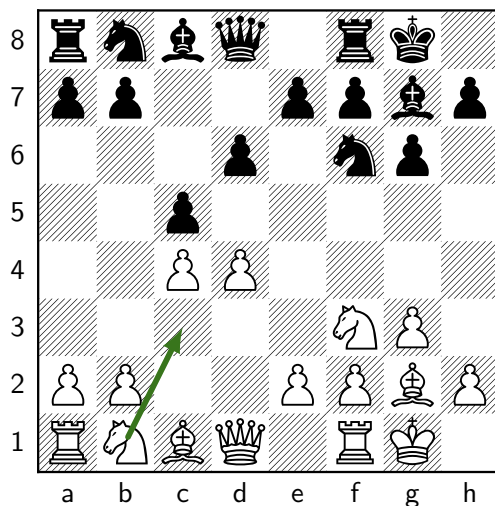
(a)



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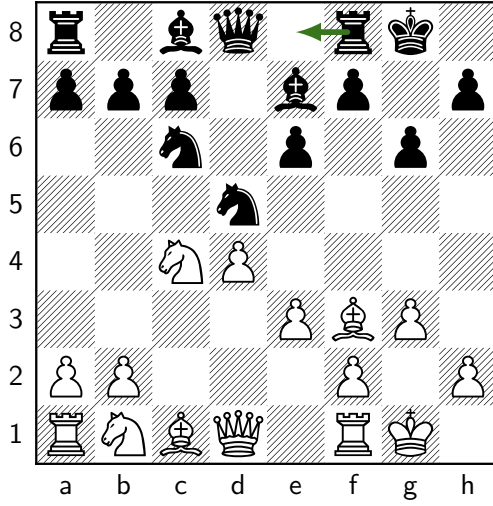


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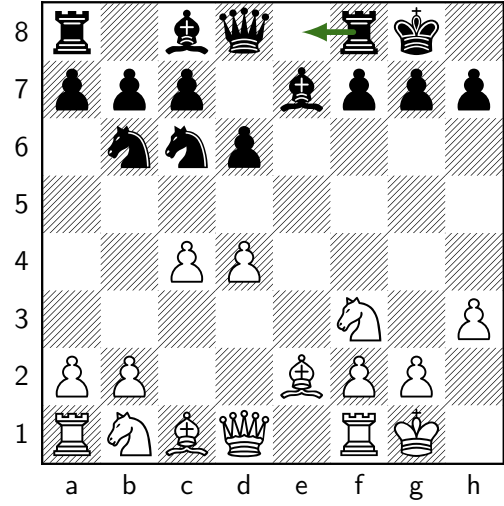


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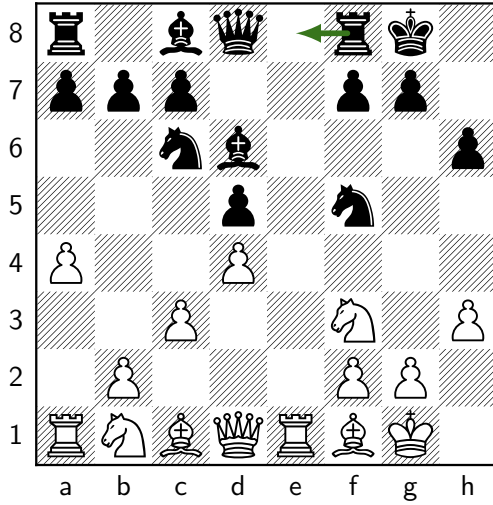
Figure 3: **Move 9, Black: Nc6**. (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3335` (ply 12) [structural distance $d=55$, QAP obj=23]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our pg3 as their Bc1, our Pf7 as their Qd1; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3336` (ply 12) [structural distance $d=55$, QAP obj=23]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our pg3 as their Bc1, our Pf7 as their Qd1; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1614` (ply 12) [structural distance $d=55$, QAP obj=23]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our pg3 as their Bc1, our Pf7 as their Qd1; transferred move b8c6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nc6 (summed QAP objective 163, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



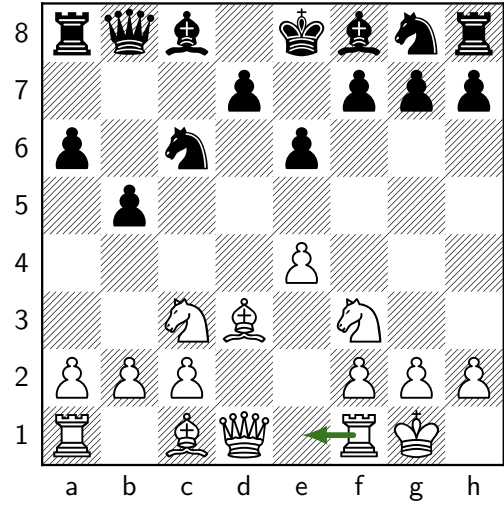
(a)



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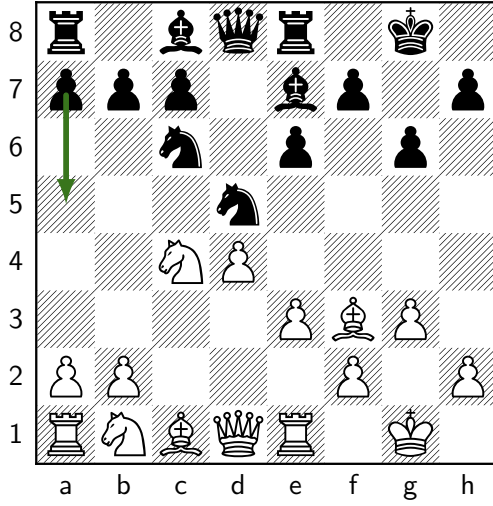


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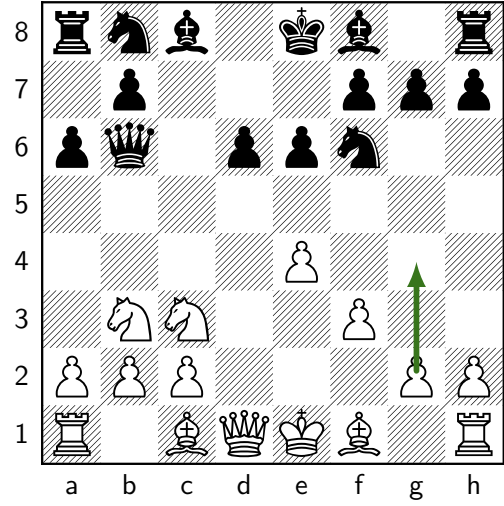


(d)

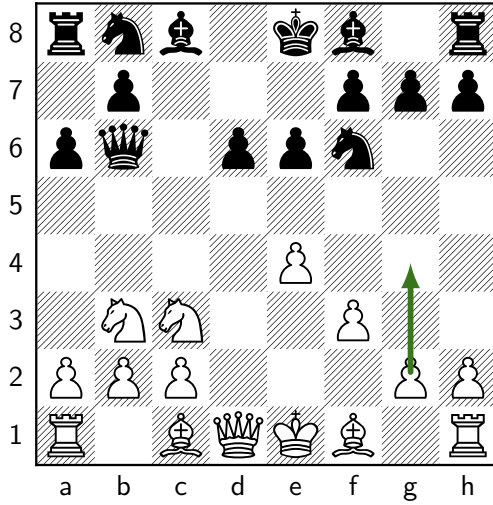
Figure 4: **Move 10, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2953` (ply 17) [structural distance $d=65$, QAP obj=22]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Qd8* as their *Ra8*, our *ra1* as their *Qd8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1293` (ply 23) [structural distance $d=56$, QAP obj=20]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *bf3* as their *Ra8*, our *Be7* as their *Qd8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#274` (ply 16) [structural distance $d=61$, QAP obj=17]: the master played Re1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pb2* as their *Ra1*, our *Nc6* as their *Bd3*; transferred move f8e8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re8 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



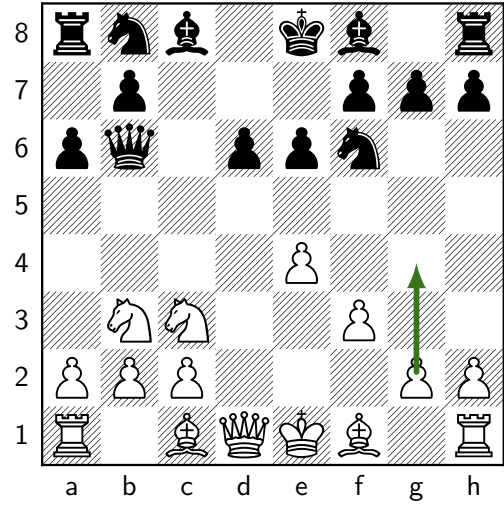
(a)



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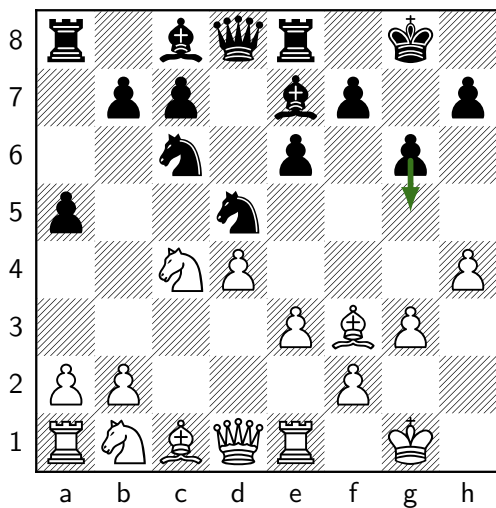


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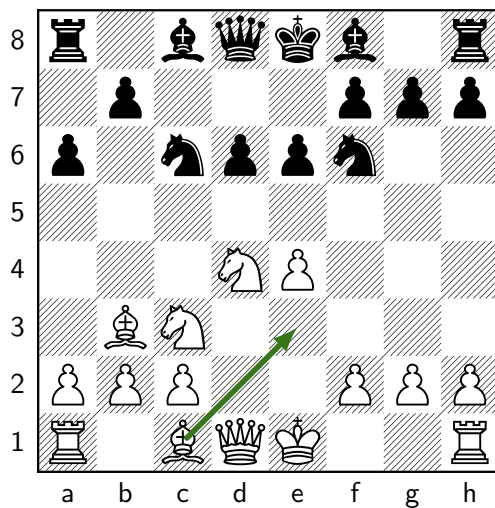


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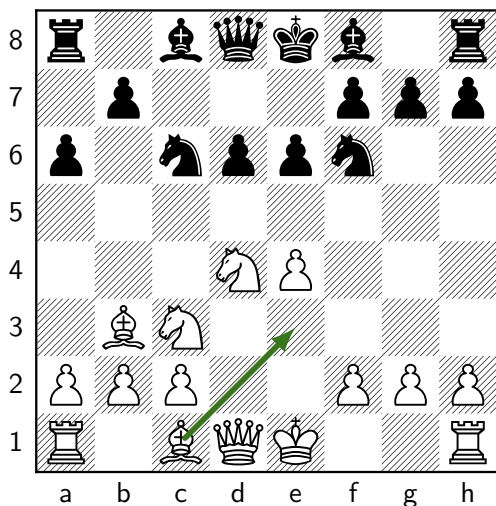
Figure 5: **Move 11, Black: a5.** (a) Our position; Melanie plays a5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#95` (ply 14) [structural distance $d=64$, QAP obj=23]: the master played g4; under μ our *Pa2* plays the role of that game’s *Pg2*; cross-type roles: our *Be7* as their *Ra1*, our *Pc7* as their *Bc1*; transferred move a7a5 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#96` (ply 14) [structural distance $d=64$, QAP obj=23]: the master played g4; under μ our *Pa2* plays the role of that game’s *Pg2*; cross-type roles: our *Be7* as their *Ra1*, our *Pc7* as their *Bc1*; transferred move a7a5 (**endorses this move**). (d) Retrieved from `ccr1_top.p0000#2906` (ply 14) [structural distance $d=64$, QAP obj=23]: the master played g4; under μ our *Pa2* plays the role of that game’s *Pg2*; cross-type roles: our *Be7* as their *Ra1*, our *Pc7* as their *Bc1*; transferred move a7a5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to a5 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



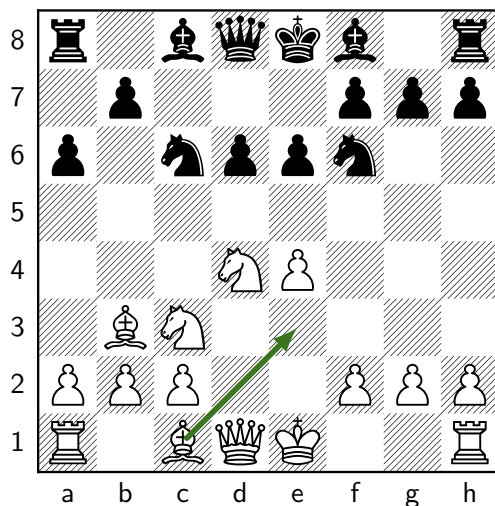
(a)



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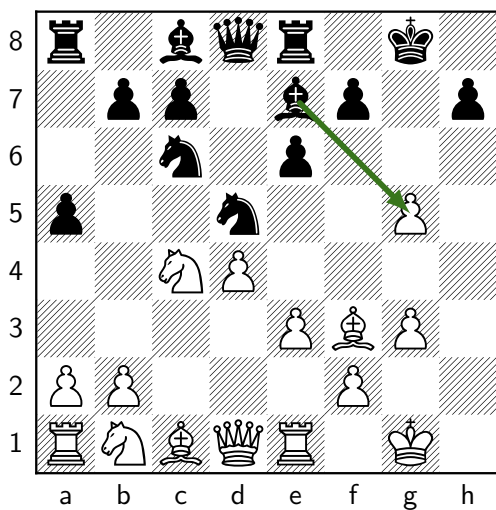


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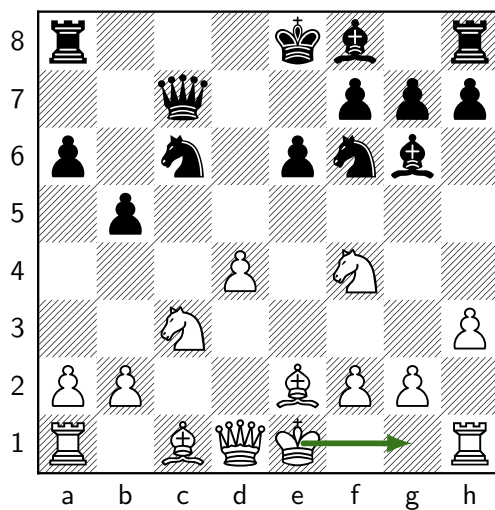


(d)

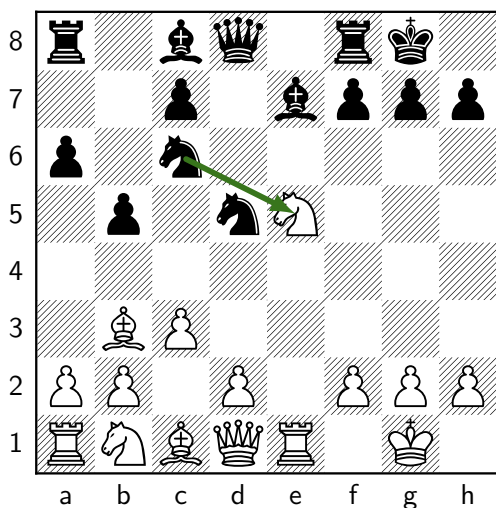
Figure 6: **Move 12, Black: g5**. (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top.p0002#41 (ply 14) [structural distance $d=69$, QAP obj=36]: the master played Be3; under μ our Pg3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra1, our pb2 as their Qd1; transferred move g6g5 (**endorses this move**). (c) Retrieved from ccr1_top.p0002#42 (ply 14) [structural distance $d=69$, QAP obj=36]: the master played Be3; under μ our Pg3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra1, our pb2 as their Qd1; transferred move g6g5 (**endorses this move**). (d) Retrieved from ccr1_top.p0002#3396 (ply 14) [structural distance $d=69$, QAP obj=36]: the master played Be3; under μ our Pg3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra1, our pb2 as their Qd1; transferred move g6g5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to g5 (summed QAP objective 335, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



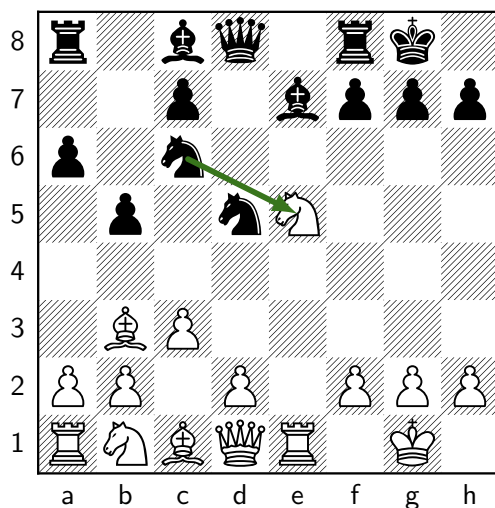
(a)



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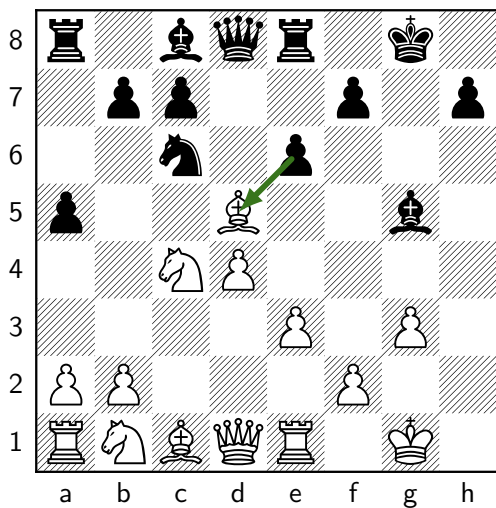


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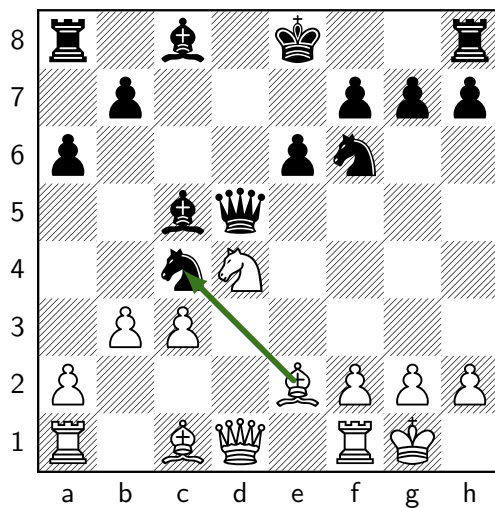


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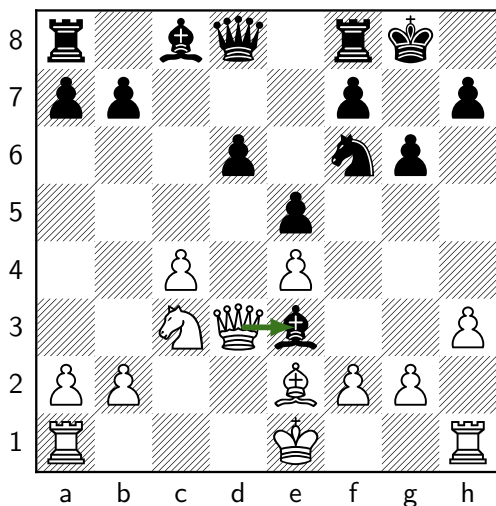
Figure 7: **Move 13, Black: Bxg5.** (a) Our position; Melanie plays Bxg5 (e7g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3671` (ply 32) [structural distance $d=72$, QAP obj=63]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra1*, our *Qd8* as their *Bc1*; transferred move g8g7. (c) Retrieved from `ccr1_top_p0000#414` (ply 19) [structural distance $d=60$, QAP obj=50]: the master played Nxe5; under μ our *Nd4* plays the role of that game’s *Nc3*; cross-type roles: our *nc4* as their *Bc8*, our *Nc6* as their *Rf8*; transferred move d5f4. (d) Retrieved from `ccr1_top_p0000#415` (ply 19) [structural distance $d=60$, QAP obj=50]: the master played Nxe5; under μ our *Nd4* plays the role of that game’s *Nc3*; cross-type roles: our *nc4* as their *Bc8*, our *Nc6* as their *Rf8*; transferred move d5f4. Synthesis: no retrieved analog transferred to Bxg5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



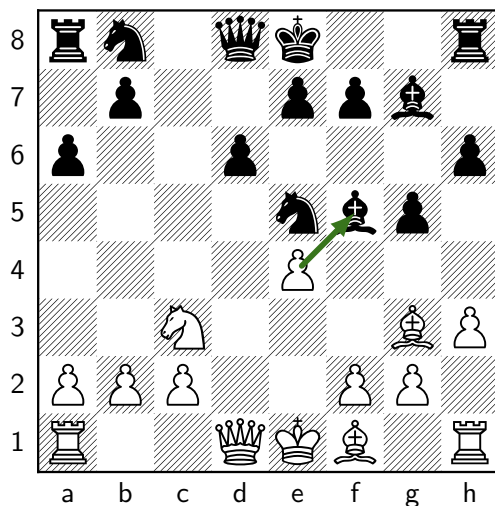
(a)



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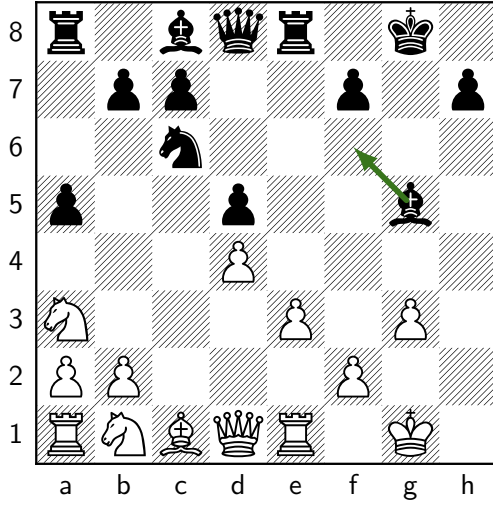


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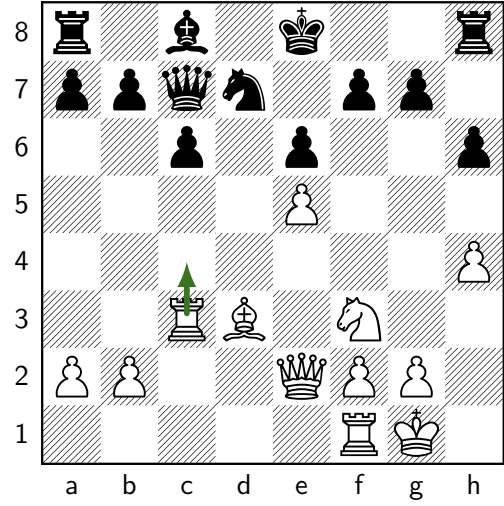


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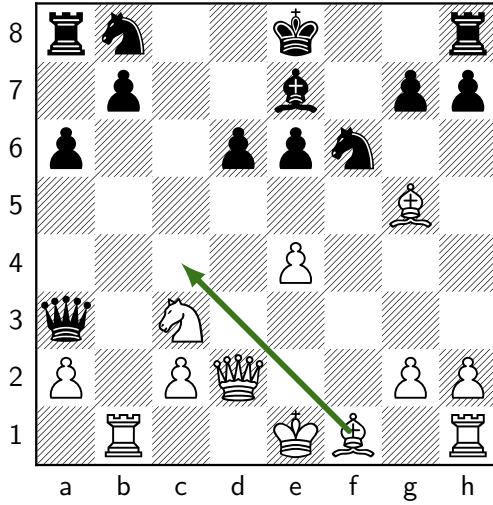
Figure 8: **Move 14, Black: exd5.** (a) Our position; Melanie plays exd5 (e6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2270` (ply 22) [structural distance $d=77$, QAP obj=65]: the master played Bxc4; under μ our *Pe3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Bg5* as their *Ra1*, our *qd1* as their *Bc1*; transferred move e6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3031` (ply 24) [structural distance $d=74$, QAP obj=44]: the master played Qxe3; under μ our *Pe3* plays the role of that game’s *Qd3* (a **cross-type** correspondence); cross-type roles: our *Pb7* as their *Rh1*, our *nc4* as their *Be2*; transferred move e6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1782` (ply 22) [structural distance $d=72$, QAP obj=43]: the master played exf5; under μ our *Pe3* plays the role of that game’s *Pe4*; cross-type roles: our *Bc8* as their *Ra1*, our *Bg5* as their *Qd1*; transferred move e6d5 (**endorses this move**). Synthesis: 29 retrieved analogs transferred to exd5 (summed QAP objective 1267, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



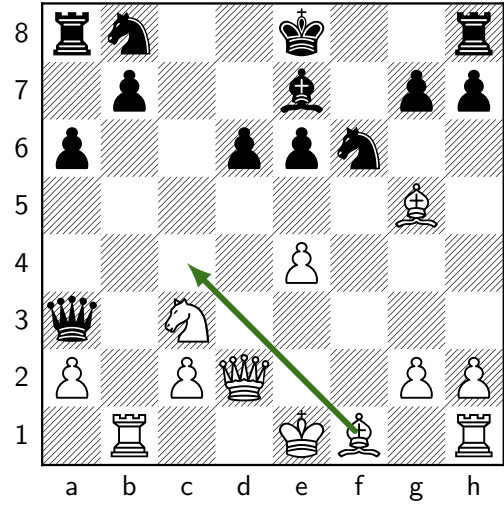
(a)



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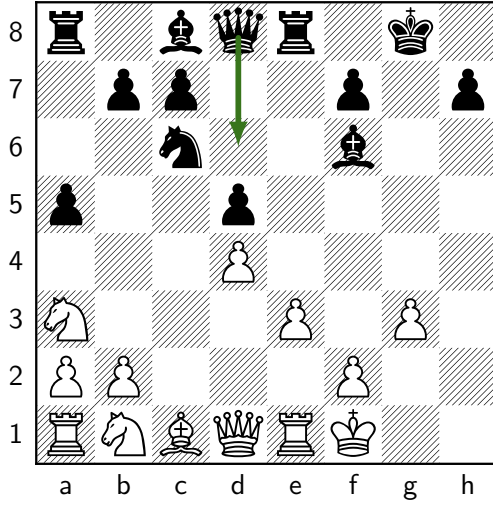


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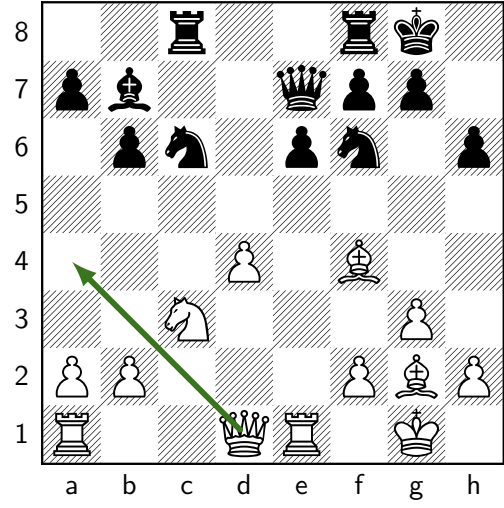


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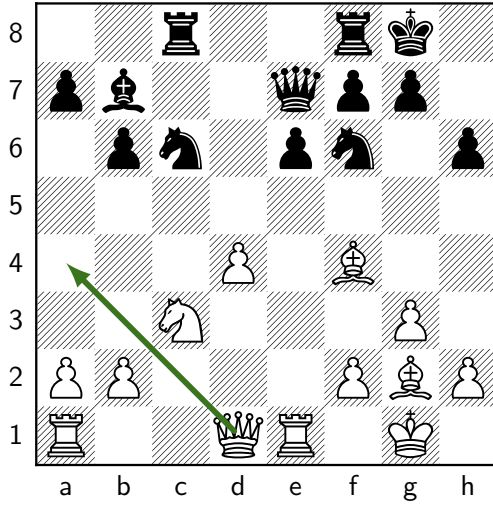
Figure 9: **Move 15, Black: Bf6.** (a) Our position; Melanie plays Bf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#611` (ply 30) [structural distance $d=83$, QAP obj=66]: the master played Rc4; under μ our Bg4 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pa2, our Qd8 as their Bd3; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#182` (ply 24) [structural distance $d=76$, QAP obj=16]: the master played Bc4; under μ our Bg4 plays the role of that game’s Bf1; cross-type roles: our Bc8 as their Pa2, our Nc6 as their Qd2; transferred move g5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#183` (ply 24) [structural distance $d=76$, QAP obj=16]: the master played Bc4; under μ our Bg4 plays the role of that game’s Bf1; cross-type roles: our Bc8 as their Pa2, our Nc6 as their Qd2; transferred move g5f6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bf6 (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



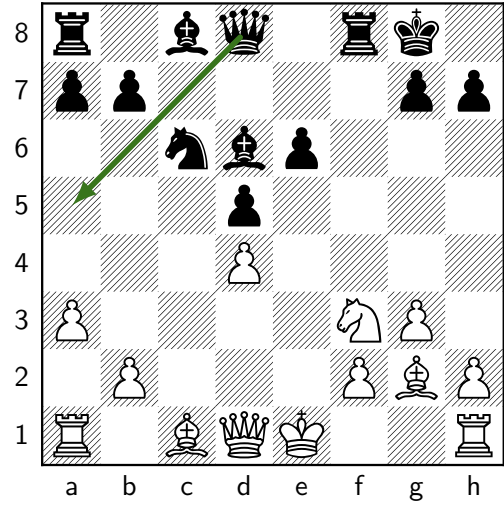
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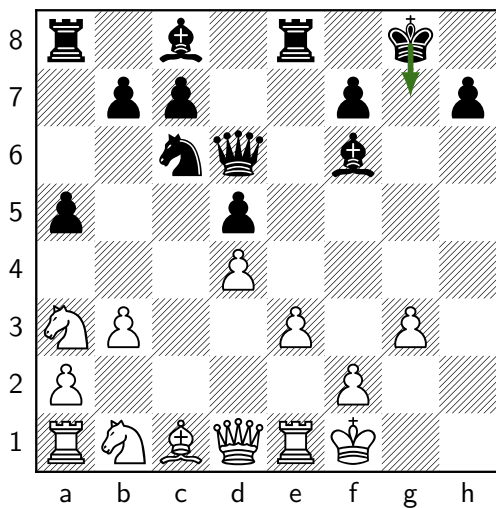


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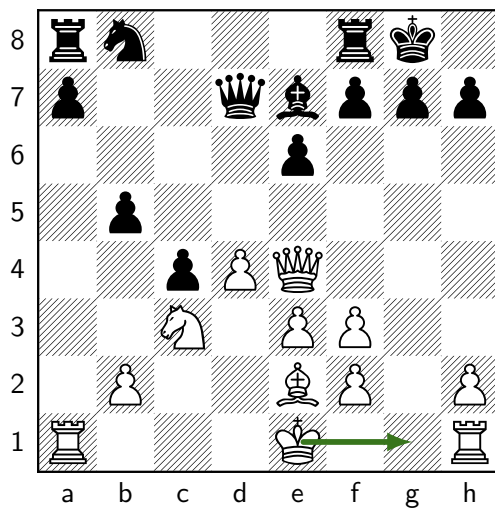


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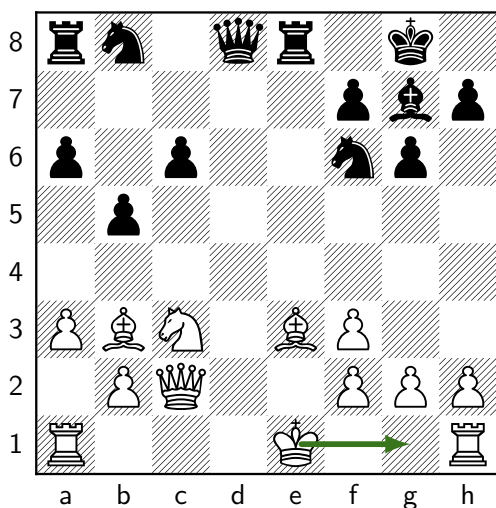
Figure 10: **Move 16, Black: Qd6.** (a) Our position; Melanie plays Qd6 (d8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#670` (ply 32) [structural distance $d=82$, QAP obj=64]: the master played Qa4; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our na3 as their Ra1, our Ra8 as their Pa2; transferred move d8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3064` (ply 32) [structural distance $d=82$, QAP obj=64]: the master played Qa4; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our na3 as their Ra1, our Ra8 as their Pa2; transferred move d8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2245` (ply 25) [structural distance $d=74$, QAP obj=45]: the master played Qa5+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pd5 as their Ra8, our na3 as their Bc8; transferred move d8d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd6 (summed QAP objective 189, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



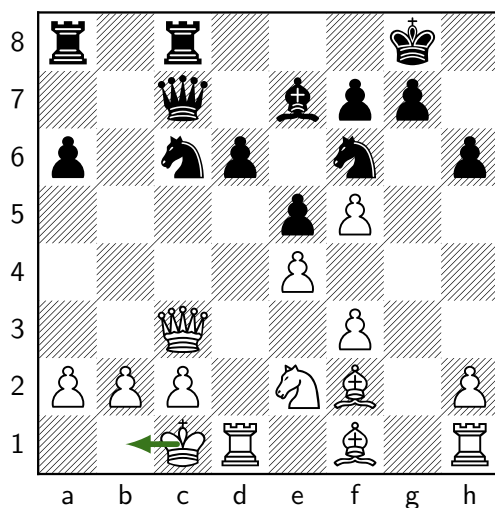
(a)



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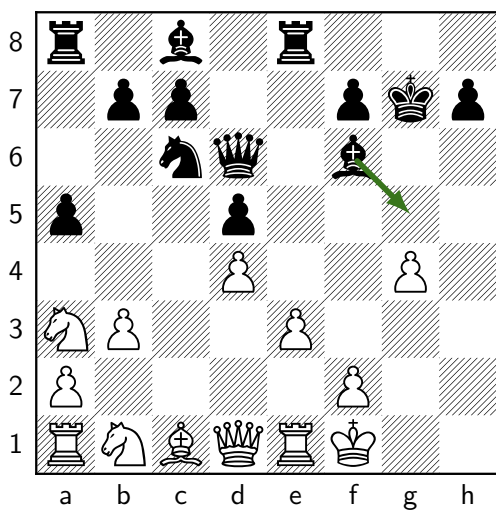


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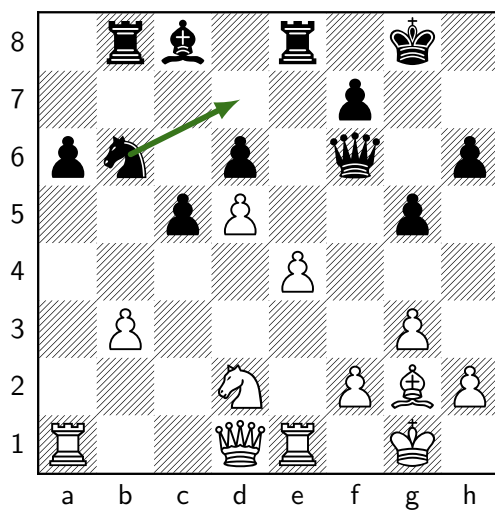


(d)

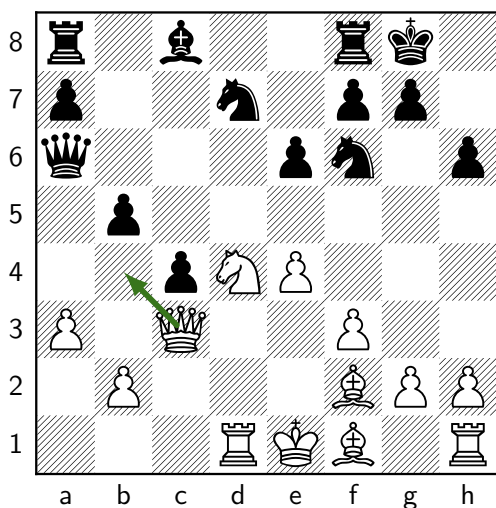
Figure 11: **Move 17, Black: Kg7**. (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2174` (ply 30) [structural distance $d=89$, QAP obj=58]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nc6 as their Be2, our Ra8 as their Pf2; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#142` (ply 28) [structural distance $d=78$, QAP obj=47]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Bc8 as their Rh1, our Re8 as their Bb3; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1292` (ply 32) [structural distance $d=82$, QAP obj=46]: the master played Kb1; under μ our Kg1 plays the role of that game’s Kc1; cross-type roles: our pa2 as their Bf1, our Bc8 as their Rh1; transferred move g8g7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg7 (summed QAP objective 283, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



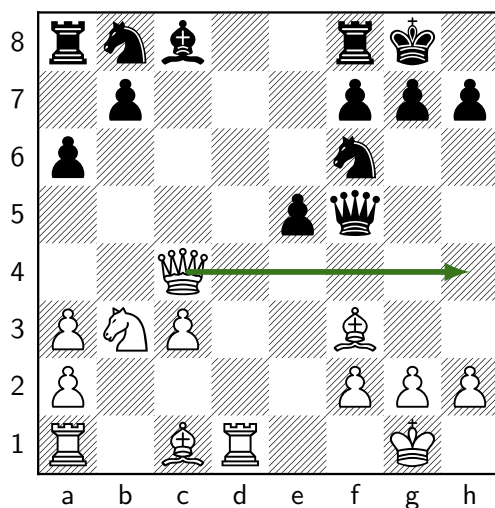
(a)



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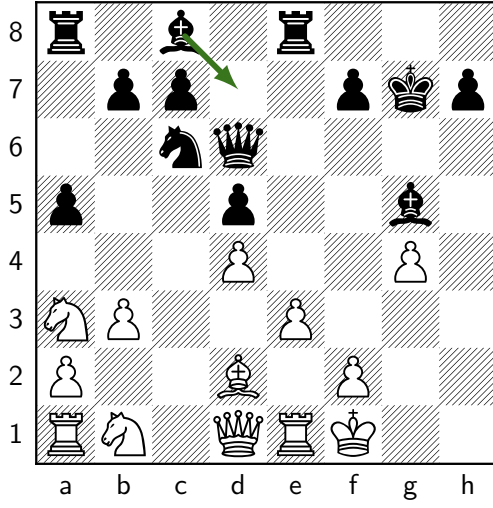


(c)

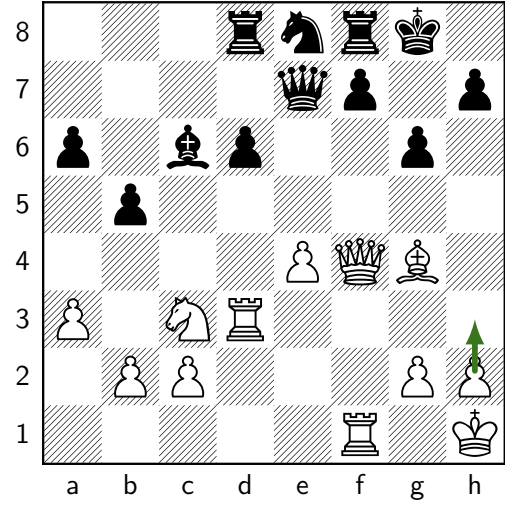


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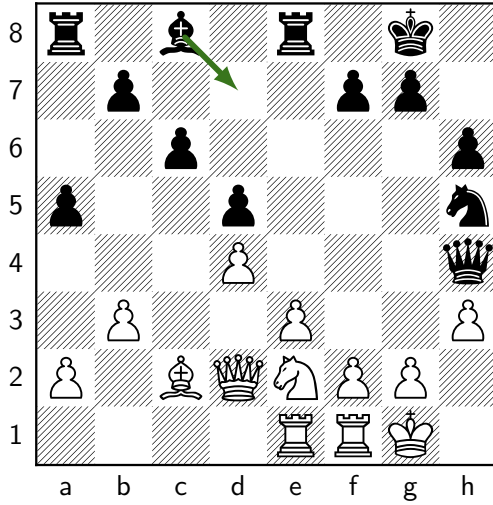
Figure 12: **Move 18, Black: Bg5**. (a) Our position; Melanie plays Bg5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2453` (ply 41) [structural distance $d=87$, QAP obj=50]: the master played Nd7; under μ our Bf3 plays the role of that game’s Nb3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rb8, our Nc6 as their Pf7; transferred move f6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1819` (ply 32) [structural distance $d=97$, QAP obj=50]: the master played Qb4; under μ our Bf3 plays the role of that game’s Qc3 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Bf1, our Pc7 as their Rh1; transferred move f6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#132` (ply 30) [structural distance $d=88$, QAP obj=41]: the master played Qh4; under μ our Bf3 plays the role of that game’s Qc4 (a **cross-type** correspondence); cross-type roles: our Re8 as their Bc1, our pb3 as their Rd1; transferred move f6g5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bg5 (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



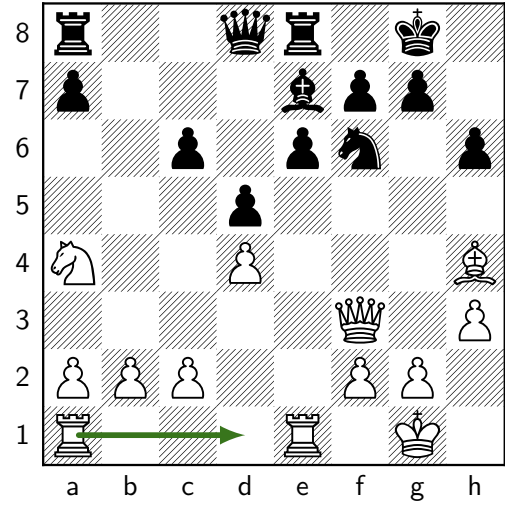
(a)



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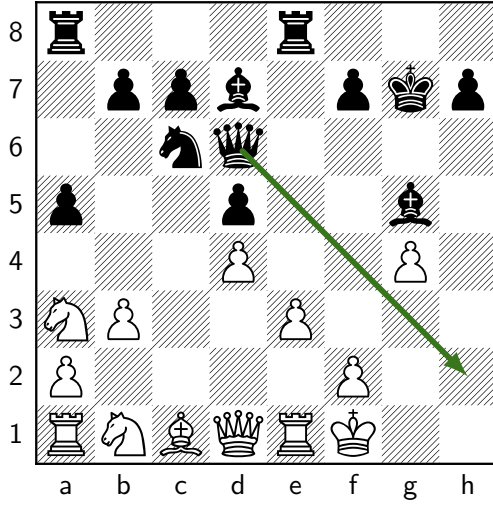


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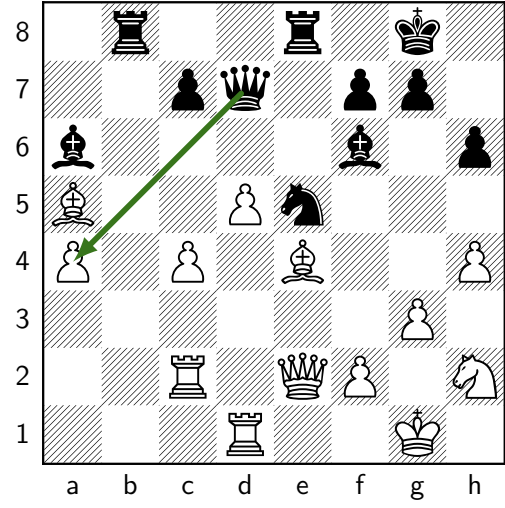


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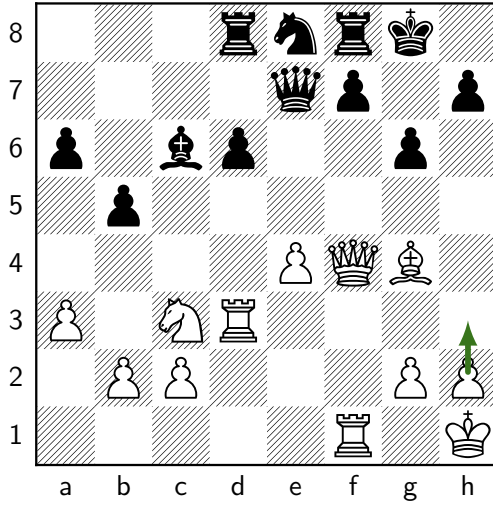
Figure 13: **Move 19, Black: Bd7 (c8d7)**. (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#451` (ply 42) [structural distance $d=72$, QAP obj=70]: the master played **h3**; under μ our *Bc1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *qd1* as their *Rf1*, our *Re8* as their *Pg2*; transferred move **c8d7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#2650` (ply 43) [structural distance $d=82$, QAP obj=52]: the master played **Bd7**; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *nb1* as their *Pg7*, our *Nc6* as their *Pc6*; transferred move **c8d7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3396` (ply 28) [structural distance $d=88$, QAP obj=39]: the master played **Rad1**; under μ our *Bc1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bg5* as their *Re1*, our *Re8* as their *Pa2*; transferred move **c8d7 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to Bd7 (summed QAP objective 161, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



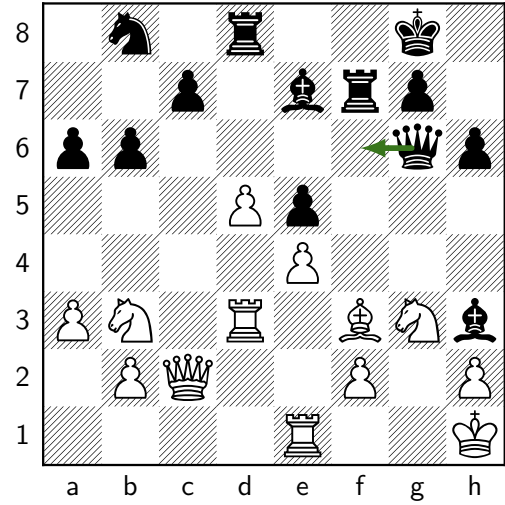
(a)



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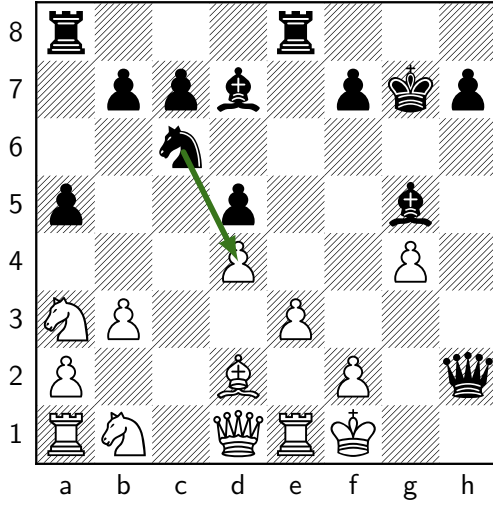


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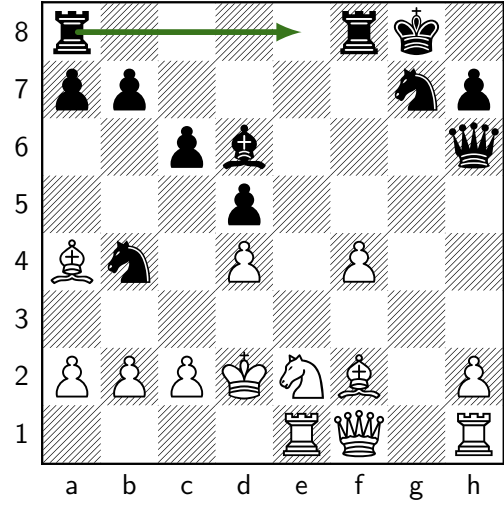


(d)

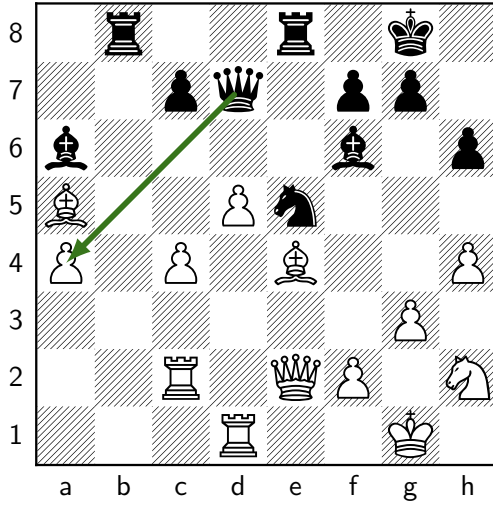
Figure 14: **Move 20, Black: Qh2.** (a) Our position; Melanie plays Qh2 (d6h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1752` (ply 47) [structural distance $d=86$, QAP obj=80]: the master played Qxa4; under μ our Bg4 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Re8, our na3 as their Pc7; transferred move g5h4. (c) Retrieved from `ccr1_top_p0000#451` (ply 42) [structural distance $d=66$, QAP obj=72]: the master played h3; under μ our Re1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Pc2, our nb1 as their Pg2; transferred move e8b8. (d) Retrieved from `ccr1_top_p0000#697` (ply 55) [structural distance $d=80$, QAP obj=71]: the master played Qf6; under μ our Bg4 plays the role of that game’s Qg3 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Nb8, our Bg5 as their Rd8; transferred move g5e7. Synthesis: no retrieved analog transferred to Qh2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



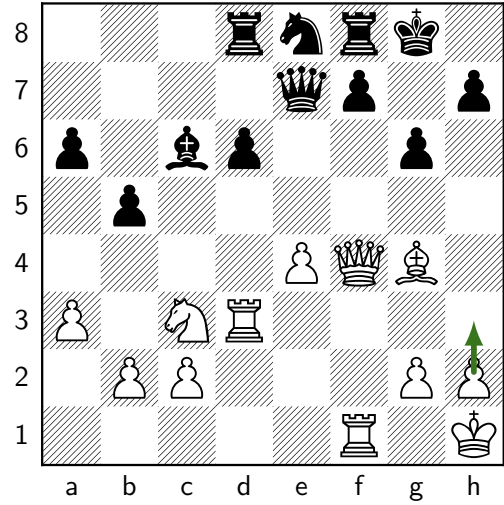
(a)



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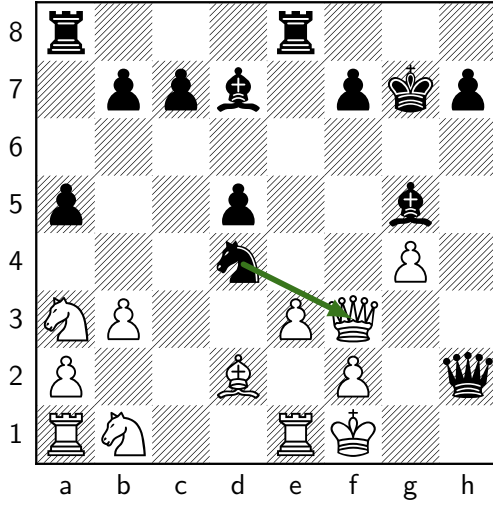


(c)

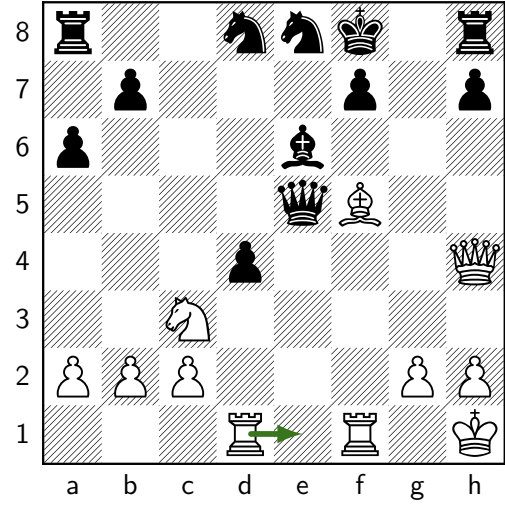


(d)

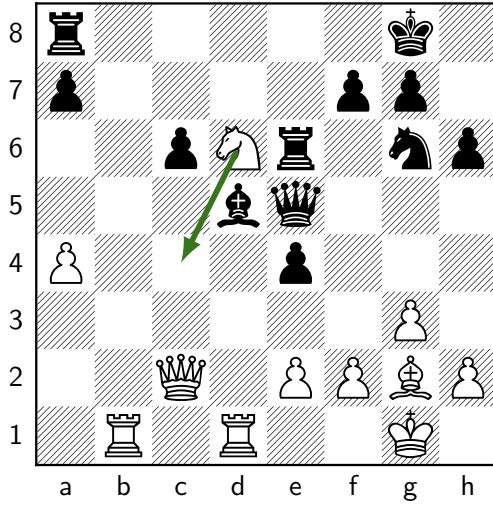
Figure 15: **Move 21, Black: Nxd4.** (a) Our position; Melanie plays Nxd4 (c6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2768` (ply 39) [structural distance $d=97$, QAP obj=140]: the master played Rae8; under μ our *Pb2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bg5* as their *Rf8*, our *ra1* as their *Pa7*; transferred move b7b6. (c) Retrieved from `ccr1_top_p0000#1752` (ply 47) [structural distance $d=92$, QAP obj=133]: the master played Qxa4; under μ our *Bg4* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *Re8*, our *Qh2* as their *Ba6*; transferred move g5h4. (d) Retrieved from `ccr1_top_p0000#451` (ply 42) [structural distance $d=90$, QAP obj=120]: the master played h3; under μ our *Re1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Bd7* as their *Rf1*, our *Bg5* as their *Rd3*; transferred move e8e7. Synthesis: no retrieved analog transferred to Nxd4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



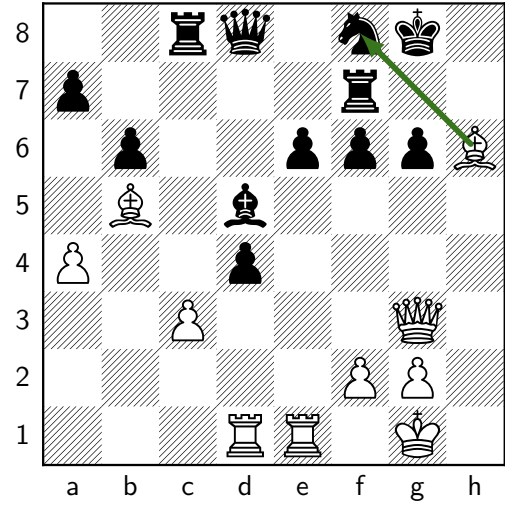
(a)



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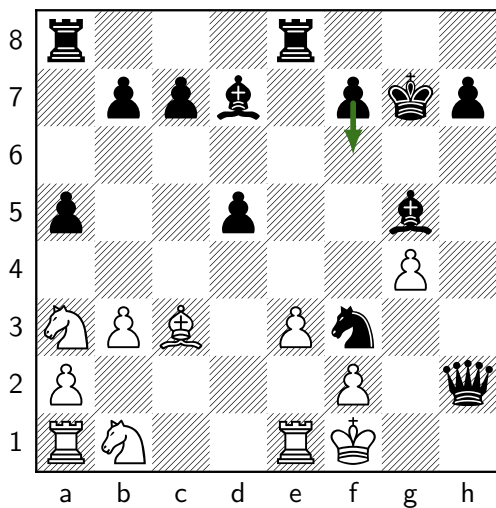


(c)

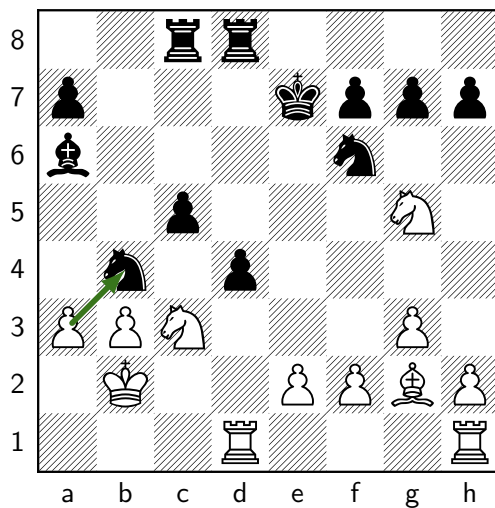


(d)

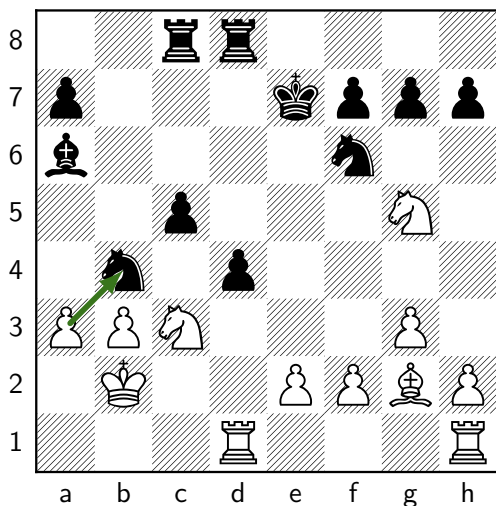
Figure 16: **Move 22, Black: Nxf3.** (a) Our position; Melanie plays Nxf3 (d4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3967` (ply 50) [structural distance $d=97$, QAP obj=78]: the master played Rde1; under μ our Bg4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Pa2, our Re8 as their Pb2; transferred move g5e7. (c) Retrieved from `ccrl_top_p0000#300` (ply 44) [structural distance $d=109$, QAP obj=65]: the master played Nc4; under μ our Nd5 plays the role of that game’s Nd6; cross-type roles: our Pd5 as their Rb1, our Pc7 as their Rd1; transferred move d4f5. (d) Retrieved from `ccrl_top_p0002#3656` (ply 54) [structural distance $d=112$, QAP obj=49]: the master played Bxf8; under μ our Nd5 plays the role of that game’s Bh6 (a **cross-type** correspondence); cross-type roles: our Bg5 as their Rd1, our Nd4 as their Pc3; transferred move d4e2. Synthesis: no retrieved analog transferred to Nxf3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



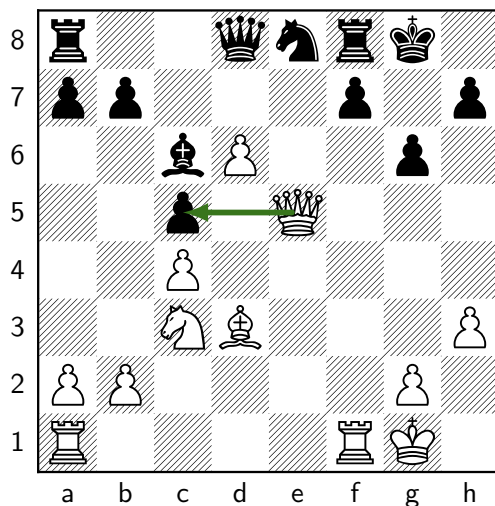
(a)



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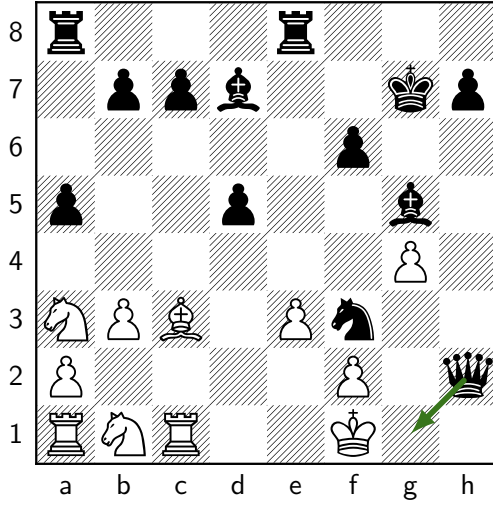


(c)

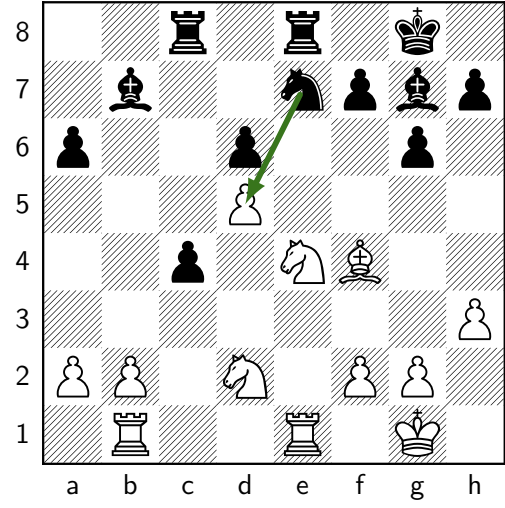


(d)

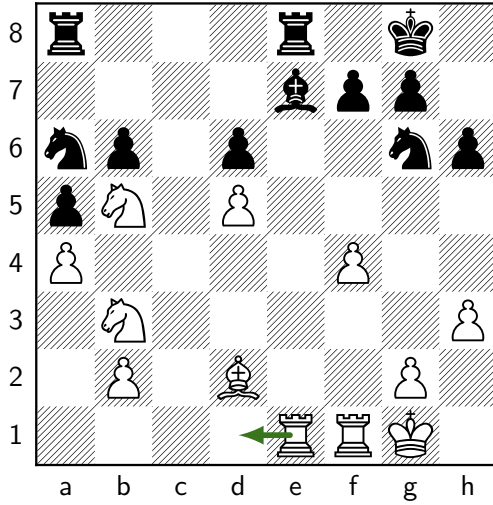
Figure 17: **Move 23, Black: f6**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1905` (ply 34) [structural distance $d=99$, QAP obj=21]: the master played **axb4**; under μ our *Pf2* plays the role of that game’s *Pa3*; cross-type roles: our *Bg5* as their *Pf2*, our *Nf3* as their *Bg2*; transferred move **f7f6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2821` (ply 34) [structural distance $d=99$, QAP obj=21]: the master played **axb4**; under μ our *Pf2* plays the role of that game’s *Pa3*; cross-type roles: our *Bg5* as their *Pf2*, our *Nf3* as their *Bg2*; transferred move **f7f6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1104` (ply 40) [structural distance $d=116$, QAP obj=12]: the master played **Qxc5**; under μ our *Pf2* plays the role of that game’s *Qe5* (a **cross-type** correspondence); cross-type roles: our *Bd7* as their *Pb2*, our *Nf3* as their *Pg2*; transferred move **f7f6** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f6 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



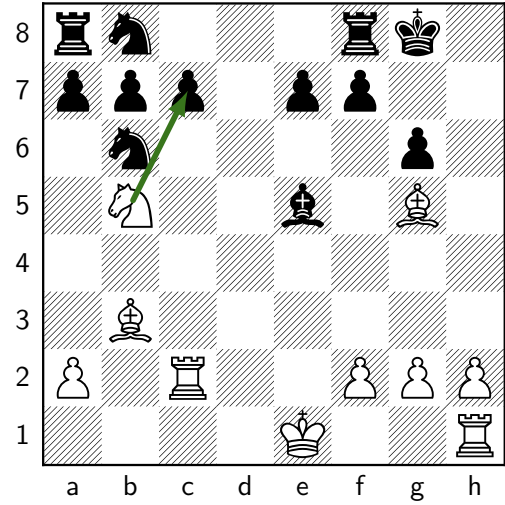
(a)



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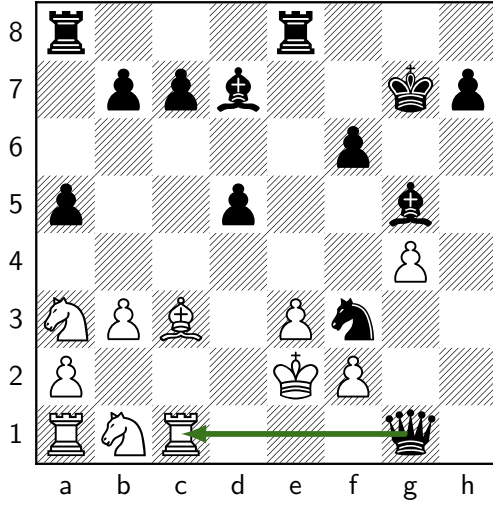


(c)

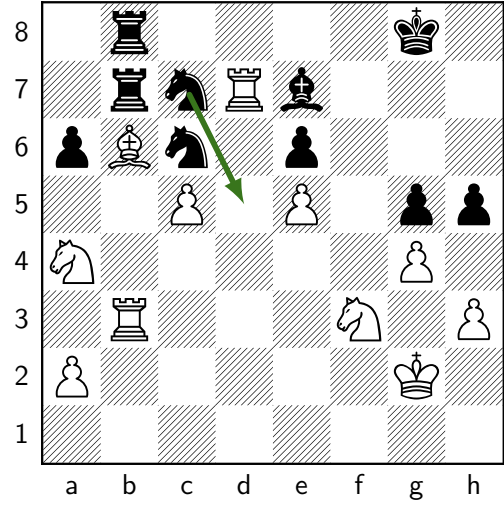


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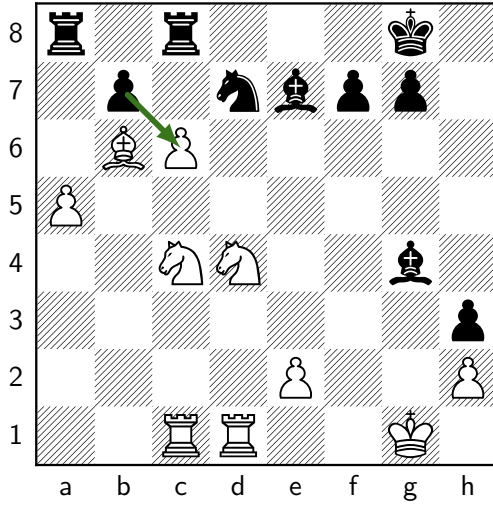
Figure 18: **Move 24, Black: Qg1+.** (a) Our position; Melanie plays Qg1+ (h2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1264` (ply 41) [structural distance $d=122$, QAP obj=117]: the master played Nxd5; under μ our Bg4 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Re8, our Re8 as their Pg6; transferred move g5h4. (c) Retrieved from `ccr1_top_p0001#3213` (ply 50) [structural distance $d=109$, QAP obj=99]: the master played Rd1; under μ our Bg4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Re8 as their Pb2, our Qh2 as their Bd2; transferred move g5h4. (d) Retrieved from `ccr1_top_p0000#136` (ply 32) [structural distance $d=107$, QAP obj=90]: the master played Nxc7; under μ our Bg4 plays the role of that game’s Nb5 (a **cross-type** correspondence); cross-type roles: our nb1 as their Rh1, our ra1 as their Pa2; transferred move g5e3. Synthesis: no retrieved analog transferred to Qg1+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



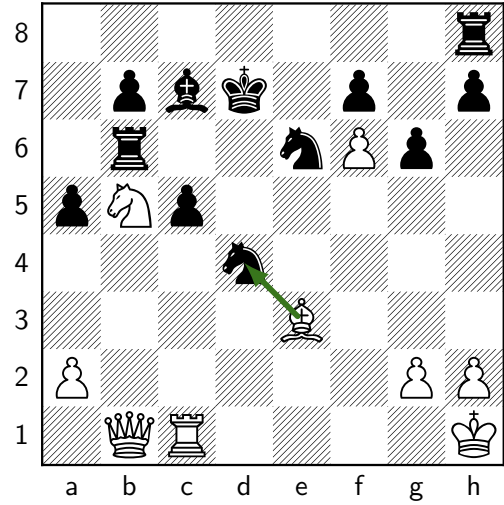
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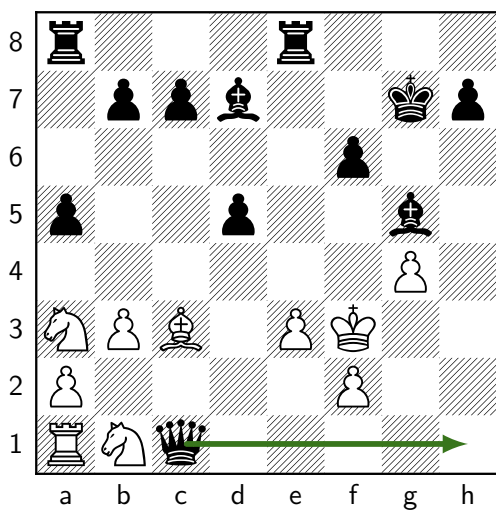


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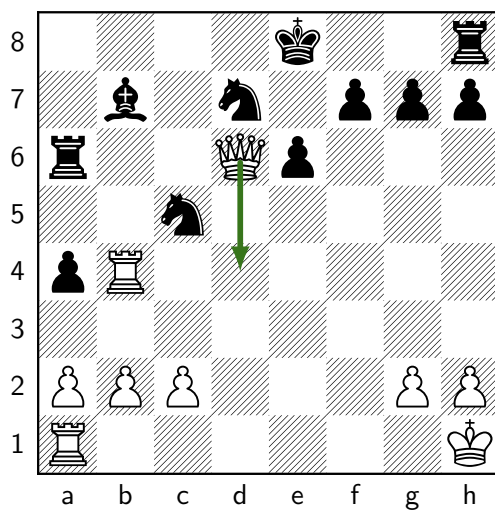


(d)

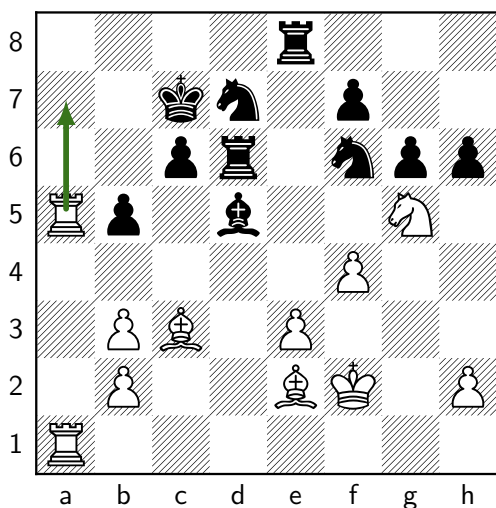
Figure 19: **Move 25, Black: Qxc1**. (a) Our position; Melanie plays Qxc1 (g1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1318` (ply 65) [structural distance $d=131$, QAP obj=59]: the master played Nd5; under μ our Nf6 plays the role of that game’s Nc2; cross-type roles: our Bg5 as their Rb7, our ke2 as their rd7; transferred move f3e5. (c) Retrieved from `ccr1_top_p0003#685` (ply 57) [structural distance $d=137$, QAP obj=50]: the master played bxc6; under μ our Qg8 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Ra8, our Bd7 as their Rc8; transferred move g1h2. (d) Retrieved from `ccr1_top_p0000#3825` (ply 58) [structural distance $d=134$, QAP obj=47]: the master played Bxd4; under μ our Qg8 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Qb1, our bc3 as their Rc1; transferred move g1g3. Synthesis: no retrieved analog transferred to Qxc1 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



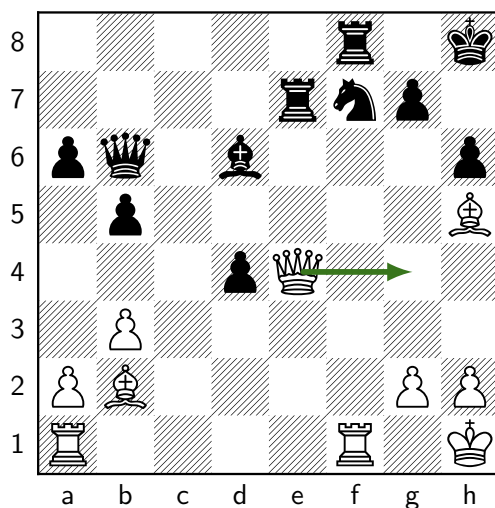
(a)



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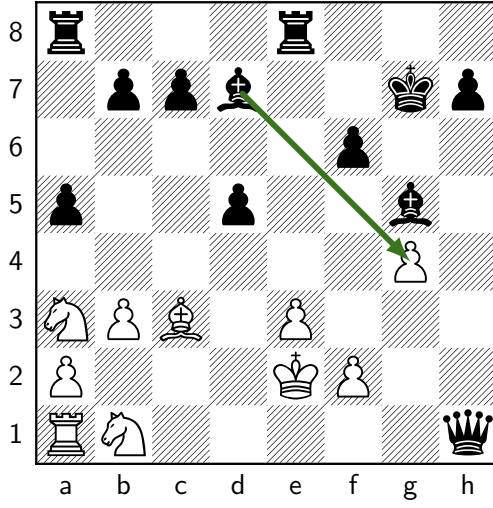


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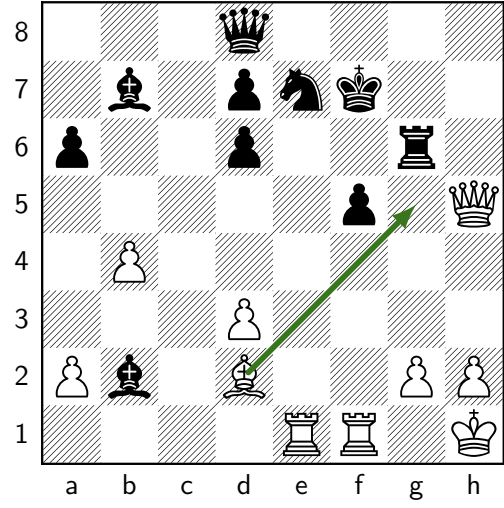


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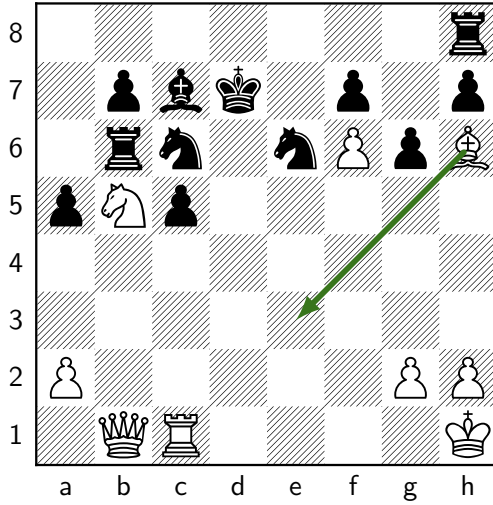
Figure 20: **Move 26, Black: Qh1+.** (a) Our position; Melanie plays Qh1+ (c1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3622` (ply 40) [structural distance $d=125$, QAP obj=81]: the master played Qd4; under μ our Qc8 plays the role of that game’s Qd6; cross-type roles: our Pa5 as their Ra1, our Bd7 as their Pc2; transferred move c1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#487` (ply 48) [structural distance $d=122$, QAP obj=36]: the master played Ra7+; under μ our Qc8 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Be2, our Qc1 as their Bc3; transferred move c1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2` (ply 50) [structural distance $d=126$, QAP obj=26]: the master played Qg4; under μ our Qc8 plays the role of that game’s Qe4; cross-type roles: our Bd7 as their Rf1, our Ra8 as their Ph2; transferred move c1h1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qh1+ (summed QAP objective 200, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



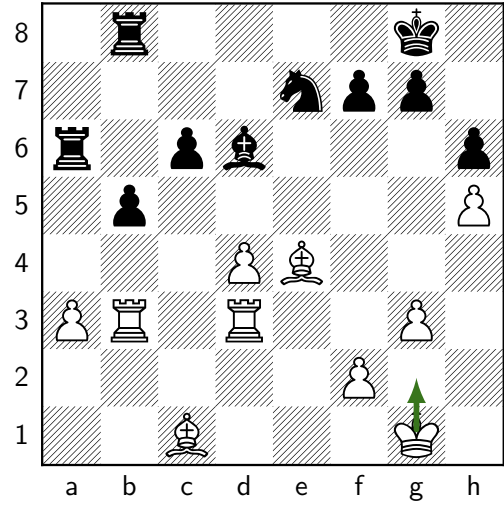
(a)



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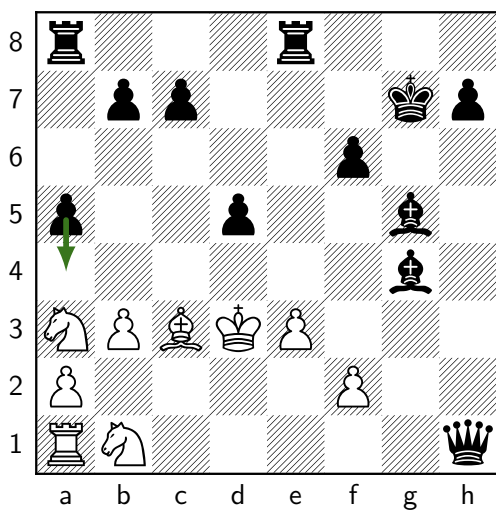


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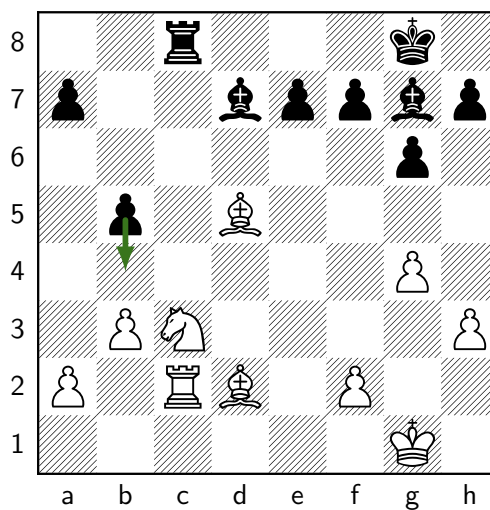


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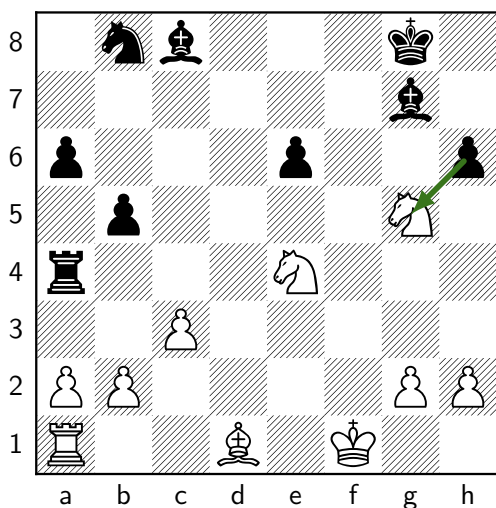
Figure 21: **Move 27, Black: Bxg4+.** (a) Our position; Melanie plays Bxg4+ (d7g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1909 (ply 50) [structural distance $d=100$, QAP obj=11]: the master played Bg5; under μ our Bd2 plays the role of that game’s Bd2; cross-type roles: our pb3 as their bb2, our na3 as their Pb4; transferred move d7g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3825 (ply 56) [structural distance $d=96$, QAP obj=9]: the master played Be3; under μ our Bd2 plays the role of that game’s Bh6; cross-type roles: our Bg5 as their Qb1, our Qh1 as their Rc1; transferred move d7g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1573 (ply 66) [structural distance $d=98$, QAP obj=104]: the master played Kg2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Re8 as their Pf2, our Qh1 as their Rb3; transferred move g7g8. Synthesis: 2 retrieved analogs transferred to Bxg4+ (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



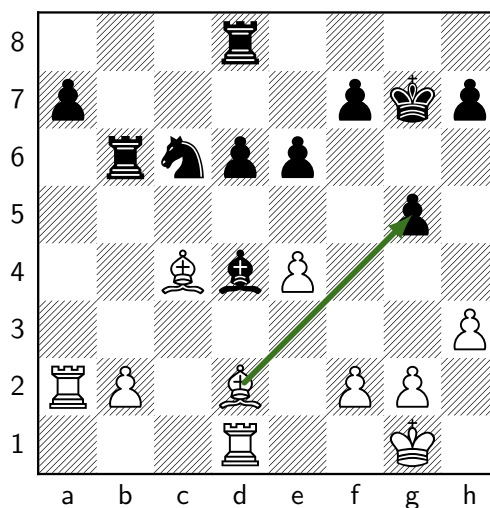
(a)



(b)

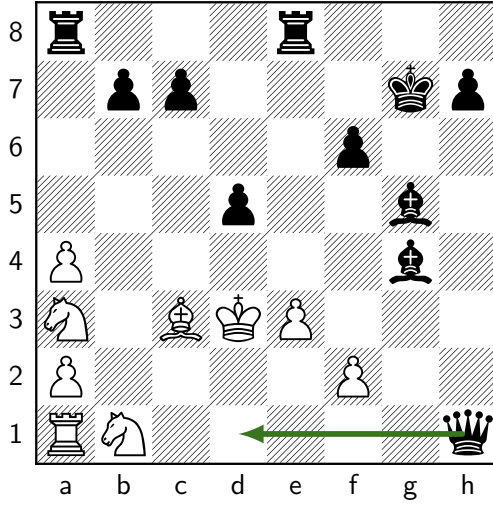


(c)

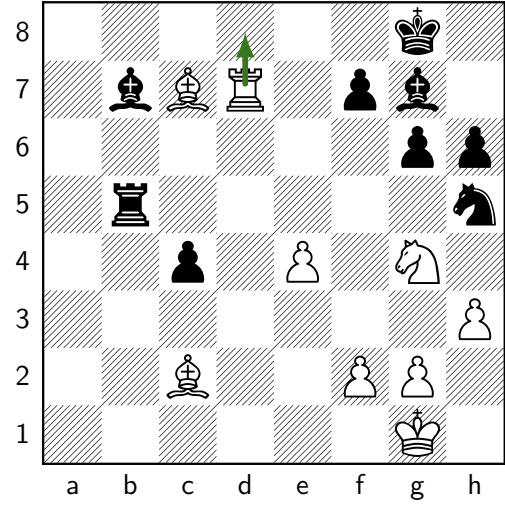


(d)

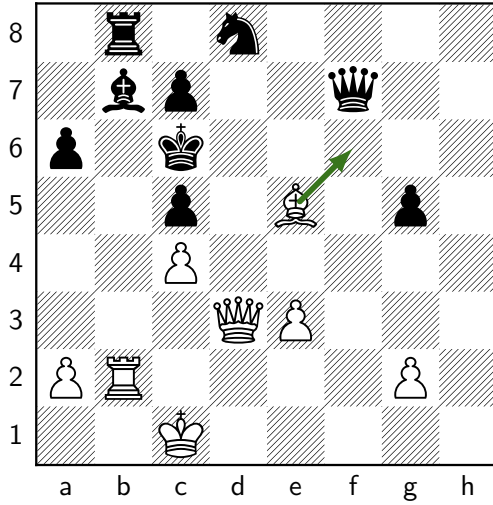
Figure 22: **Move 28, Black: a4.** (a) Our position; Melanie plays **a4** (**a5a4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#2388` (ply 49) [structural distance $d=90$, QAP obj=76]: the master played **b4**; under μ our $Pa4$ plays the role of that game’s $Pb4$; cross-type roles: our $Qh1$ as their $Rc8$, our $Re8$ as their $Pa7$; transferred move **a5a4** (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#385` (ply 43) [structural distance $d=94$, QAP obj=27]: the master played **hxg5**; under μ our $Pa4$ plays the role of that game’s $Ph3$; cross-type roles: our $Pb7$ as their $Nb8$, our $Pd5$ as their $Bg7$; transferred move **a5a4** (**endorses this move**). (d) Retrieved from `ccl1_top_p0001#3727` (ply 42) [structural distance $d=91$, QAP obj=90]: the master played **Bxg5**; under μ our $Bg4$ plays the role of that game’s $Bd2$; cross-type roles: our $nb1$ as their $Pb2$, our $nb1$ as their $Pf2$; transferred move **g5f4**. Synthesis: 2 retrieved analogs transferred to **a4** (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



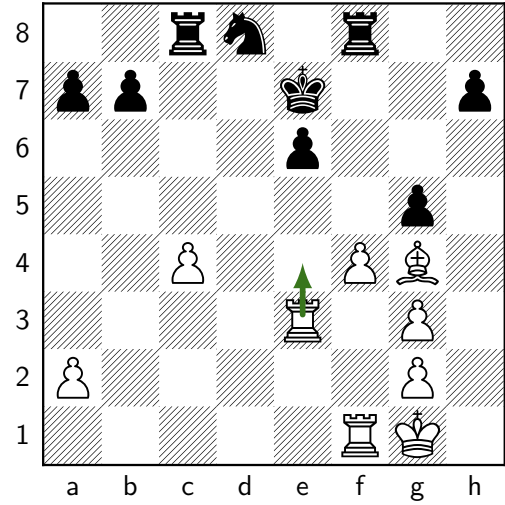
(a)



(b)

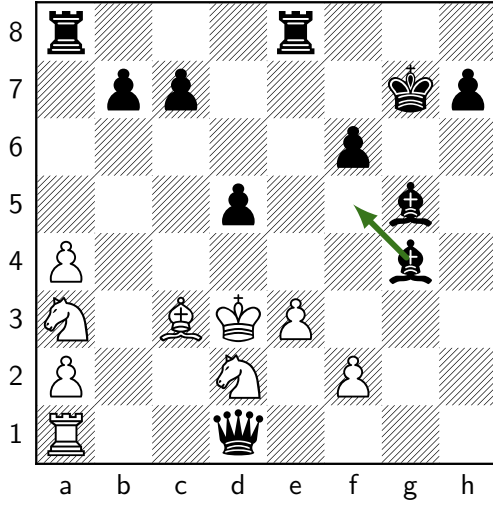


(c)

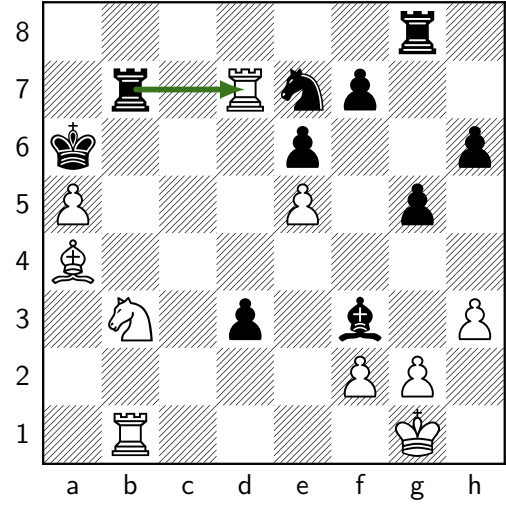


(d)

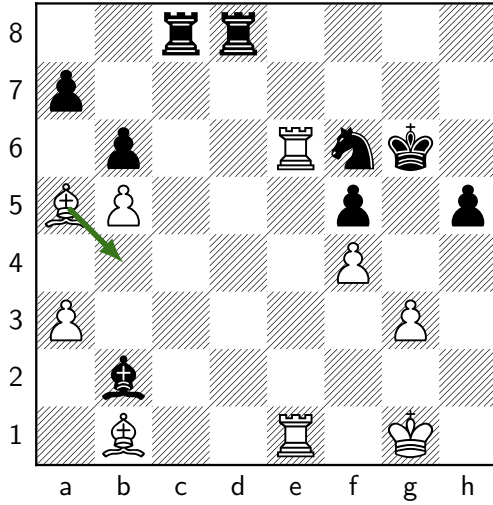
Figure 23: **Move 29, Black: Qd1+.** (a) Our position; Melanie plays Qd1+ (h1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#95` (ply 68) [structural distance $d=85$, QAP obj=71]: the master played R $d8+$; under μ our $Qh8$ plays the role of that game’s $Rd7$ (a **cross-type** correspondence); cross-type roles: our $pa2$ as their $Bc2$, our $Ra8$ as their $Ph3$; transferred move $h1d1$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1865` (ply 66) [structural distance $d=90$, QAP obj=81]: the master played B $f6$; under μ our $Qh8$ plays the role of that game’s $Be5$ (a **cross-type** correspondence); cross-type roles: our $Qh1$ as their $Rb2$, our $Bg5$ as their $Qd3$; transferred move $h1h2$. (d) Retrieved from `ccr1_top_p0002#427` (ply 48) [structural distance $d=92$, QAP obj=75]: the master played R $e4$; under μ our $Qh8$ plays the role of that game’s $Re3$ (a **cross-type** correspondence); cross-type roles: our $Bg5$ as their $Rf1$, our $Re8$ as their $Pa2$; transferred move $h1h2$. Synthesis: 2 retrieved analogs transferred to Qd1+ (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



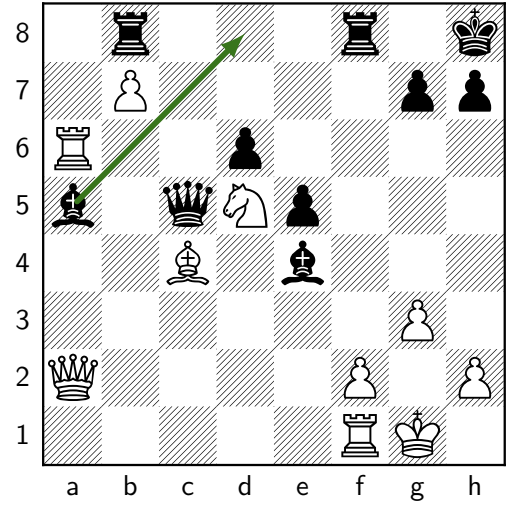
(a)



(b)

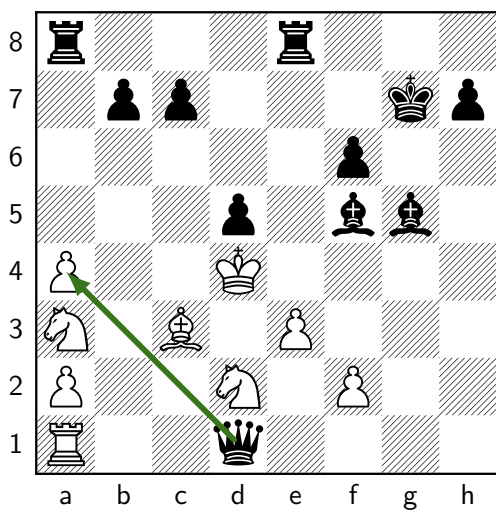


(c)

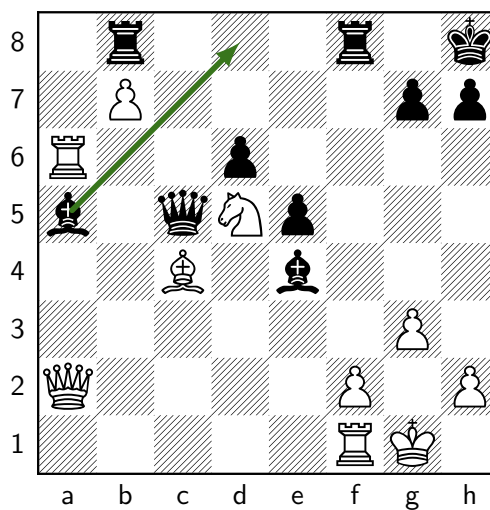


(d)

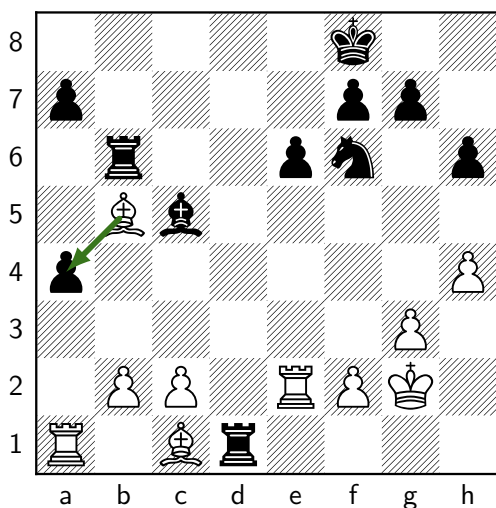
Figure 24: **Move 30, Black: Bf5+.** (a) Our position; Melanie plays Bf5+ (g4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2101` (ply 63) [structural distance $d=99$, QAP obj=87]: the master played Rxd7; under μ our Qd8 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Bg4 as their Rg8, our Qd1 as their Ne7; transferred move d1a1. (c) Retrieved from `ccr1_top_p0001#1972` (ply 74) [structural distance $d=104$, QAP obj=70]: the master played Bb4; under μ our Qd8 plays the role of that game’s Ba5 (a **cross-type** correspondence); cross-type roles: our Bg4 as their Re1, our Ra8 as their Pb5; transferred move d1a4. (d) Retrieved from `ccr1_top_p0000#2525` (ply 65) [structural distance $d=104$, QAP obj=66]: the master played Bd8; under μ our Qd8 plays the role of that game’s Ba4 (a **cross-type** correspondence); cross-type roles: our Bg5 as their Rf8, our Re8 as their Pg7; transferred move d1a4. Synthesis: no retrieved analog transferred to Bf5+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



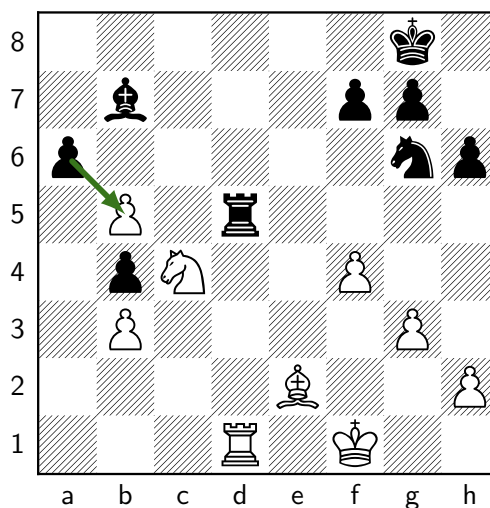
(a)



(b)

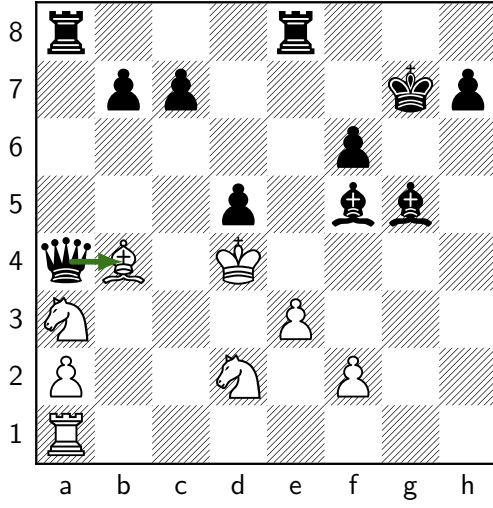


(c)

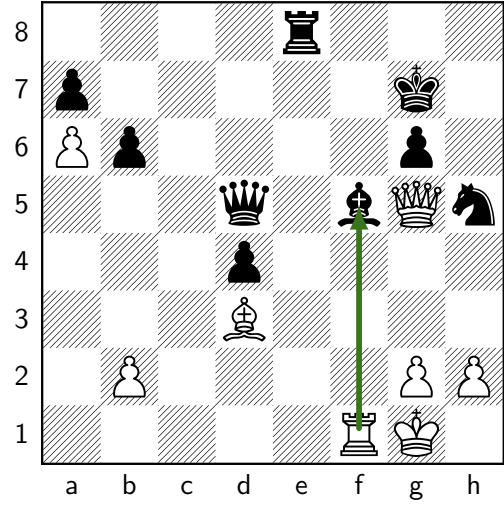


(d)

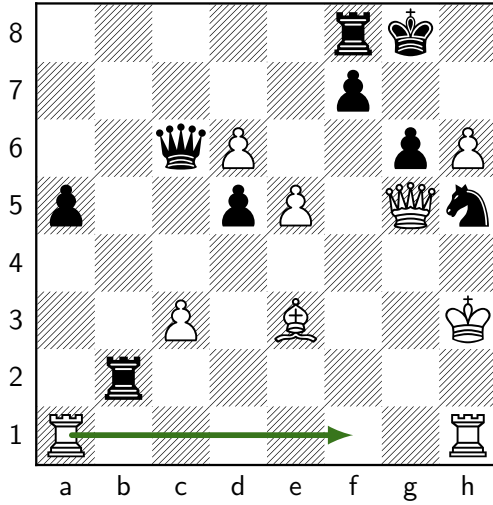
Figure 25: **Move 31, Black: Qxa4+.** (a) Our position; Melanie plays Qxa4+ (d1a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2525` (ply 65) [structural distance $d=93$, QAP obj=67]: the master played Bd8; under μ our Qd8 plays the role of that game’s Ba4 (a **cross-type** correspondence); cross-type roles: our kd4 as their Rb8, our Bg5 as their Rf8; transferred move d1a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#83` (ply 50) [structural distance $d=107$, QAP obj=56]: the master played Bxa4; under μ our Qd8 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ra1, our nd2 as their Bc1; transferred move d1a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3029` (ply 71) [structural distance $d=97$, QAP obj=24]: the master played axb5; under μ our Qd8 plays the role of that game’s Pa3 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Bb7, our Bf5 as their Ng6; transferred move d1a4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qxa4+ (summed QAP objective 231, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



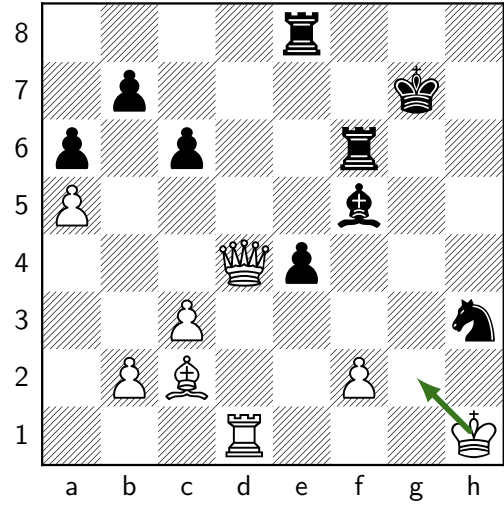
(a)



(b)

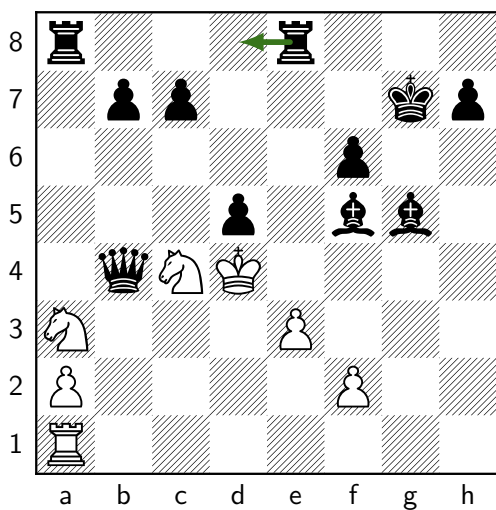


(c)

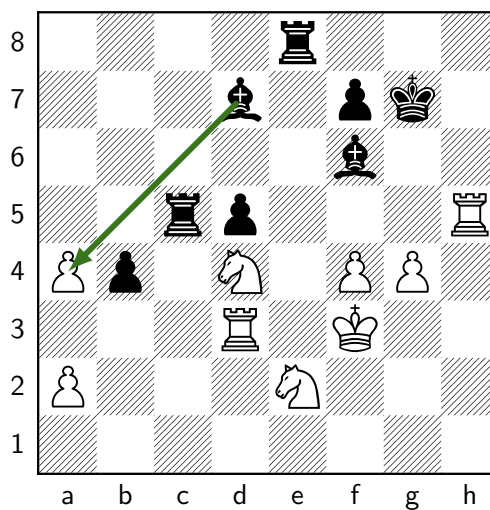


(d)

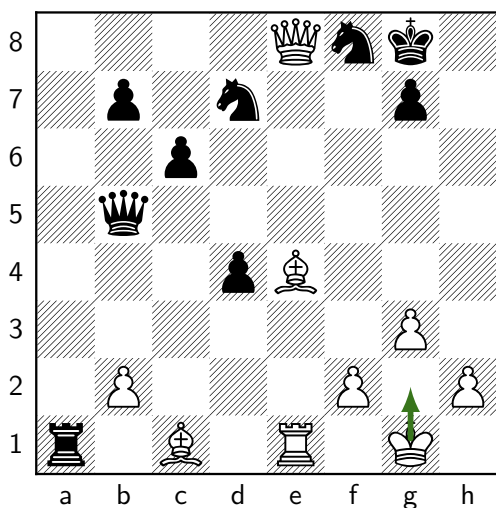
Figure 26: **Move 32, Black: Qxb4+.** (a) Our position; Melanie plays Qxb4+ (a4b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1433` (ply 62) [structural distance $d=104$, QAP obj=115]: the master played Rxf5; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *Ra8* as their *Pg2*, our *Qa4* as their *Bd3*; transferred move e8e6. (c) Retrieved from `ccr1_top_p0001#3521` (ply 60) [structural distance $d=100$, QAP obj=63]: the master played Raf1; under μ our *Pb2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pc7* as their *Rh1*, our *Ph7* as their *rb2*; transferred move b7b6. (d) Retrieved from `ccr1_top_p0001#1206` (ply 68) [structural distance $d=102$, QAP obj=56]: the master played Kg2; under μ our *Kg2* plays the role of that game’s *Kh1*; cross-type roles: our *Pd5* as their *Bc2*, our *Re8* as their *Pf2*; transferred move g7h8. Synthesis: no retrieved analog transferred to Qxb4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



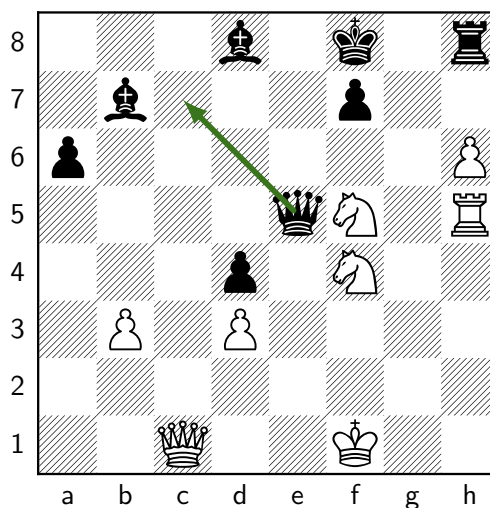
(a)



(b)

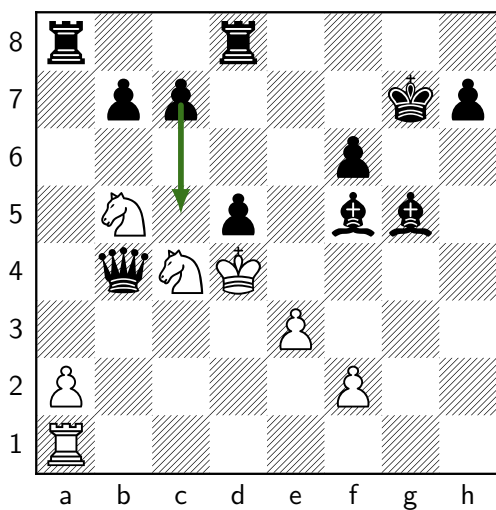


(c)

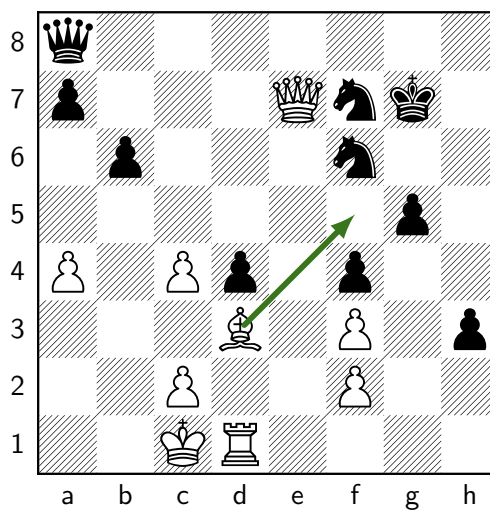


(d)

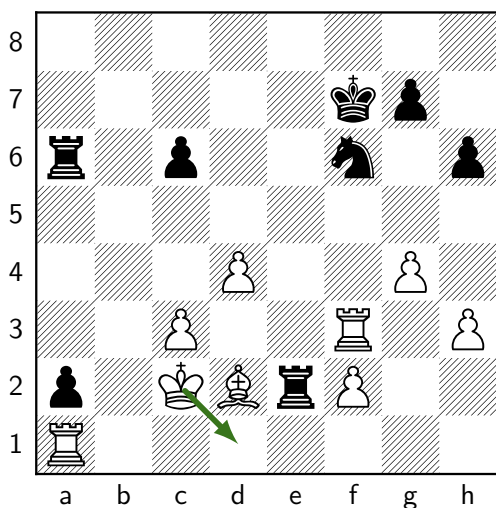
Figure 27: **Move 33, Black: Red8.** (a) Our position; Melanie plays Red8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2610` (ply 73) [structural distance $d=107$, QAP obj=12]: the master played Bxa4; under μ our *Re1* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *Pc7* as their *Re8*, our *Qb4* as their *Bf6*; transferred move e8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1638` (ply 60) [structural distance $d=102$, QAP obj=97]: the master played Kg2; under μ our *Kg2* plays the role of that game’s *Kg1*; cross-type roles: our *Qb4* as their *ra1*, our *nc4* as their *Bc1*; transferred move g7h6. (d) Retrieved from `ccr1_top_p0000#3479` (ply 67) [structural distance $d=106$, QAP obj=78]: the master played Qc7; under μ our *Qb5* plays the role of that game’s *Qe4*; cross-type roles: our *Pf6* as their *Bd8*, our *Pd5* as their *Bb7*; transferred move b4b6. Synthesis: 1 retrieved analog transferred to Red8 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



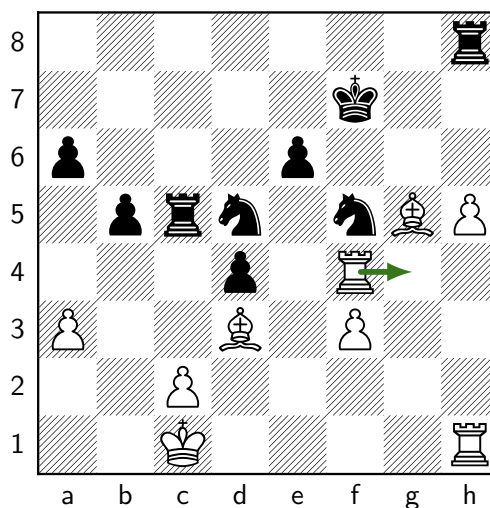
(a)



(b)



(c)



(d)

Figure 28: **Move 34, Black: c5#**. (a) Our position; Melanie plays c5# (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1482` (ply 76) [structural distance $d=92$, QAP obj=96]: the master played Bf5; under μ our Pd4 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Rd1, our ra1 as their Pc2; transferred move d5c4. (c) Retrieved from `ccrl_top_p0000#3490` (ply 76) [structural distance $d=93$, QAP obj=64]: the master played Kd1; under μ our Kg2 plays the role of that game’s Kc2; cross-type roles: our Qb4 as their Ra1, our nc4 as their pa2; transferred move g7g8. (d) Retrieved from `ccrl_top_p0001#2223` (ply 64) [structural distance $d=83$, QAP obj=63]: the master played Rg4; under μ our Qb5 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Bd3, our Bf5 as their Pf3; transferred move b4a4. Synthesis: no retrieved analog transferred to c5# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.